(b) (i) State Gauss's law in electrostatics. Using it, derive an expression for the electric field due to a uniformly charged thin spherical shell of radius R at a point (i) outside, and (ii) inside the shell.

(ii) A point charge of 4 μC is at the centre of a cubic Gaussian surface, 1·0 m on edge. Find the electric flux through one of the faces of the Gaussian surface. *5*

(b)

(i) Statement of Gauss's law in electrostatics.	1
Derivation of electric field due to a uniformly charged thin spherical shell	
(i) Outside the shell	1
(ii) Inside the shell	1
(ii) Calculation of electric flux.	2

(i) Gauss's law in electrostatics states that net electric flux linked with a closed surface equals $\frac{1}{\varepsilon_o}$ times the charge enclosed by the surface.

Consider a conducting spherical shell of radius R, charge given to it is q.

(i) For point P outside the shell construct a concentric sphere(Gaussian surface) of radius r (r>R)

Electric flux linked with the Gaussian sphere $\phi_E = E4\pi r^2$ ____________(i) 1

Charge enclosed by the Gaussian spherical surface = q

Gauss's Law

$$\phi_E = \frac{q}{\varepsilon_o}$$

½

$$E4\pi r^2 = \frac{q}{\varepsilon_o}$$

$$E = \frac{q}{4\pi\varepsilon_o r^2}$$

½

(ii) For a point inside the shell
Charge enclosed
by the Gaussian spherical surface of radius r' (r'<R)=0 ½

$$\therefore E4\pi r'^2 = \frac{q_{in}}{\varepsilon_o} = 0$$

½

$$E = 0$$

(ii) Net electric flux linked with the cube

$$\phi'_{Enet} = \frac{q}{\varepsilon_o}$$

½

Electric flux linked with one face

$$\phi'_E = \frac{\phi_{Enet}}{\varepsilon_o} = \frac{q}{6\varepsilon_o}$$

½

$$\phi'_E = \frac{4\times10^{-6}}{6\times8.854\times10^{-12}}$$

$$= 7.5\times10^4\ \mathrm{Nm^2C^{-1}}$$

1

6. A point charge q is kept at a distance r from an infinitely long straight wire with charge density λ. The magnitude of the electrostatic force experienced by charge q is :

(a) Zero

(b) $\dfrac{q\lambda}{2\pi \varepsilon_0 r}$

(c) $\dfrac{q\lambda}{4\pi \varepsilon_0 r}$

(d) $\dfrac{q\lambda}{\varepsilon_0 r}$

6. A point charge q is kept at a distance r from an infinitely long straight wire with charge density λ. The magnitude of the electrostatic force experienced by charge q is :

(a) Zero

(b) $\dfrac{q\lambda}{2\pi \varepsilon_0 r}$

(c) $\dfrac{q\lambda}{4\pi \varepsilon_0 r}$

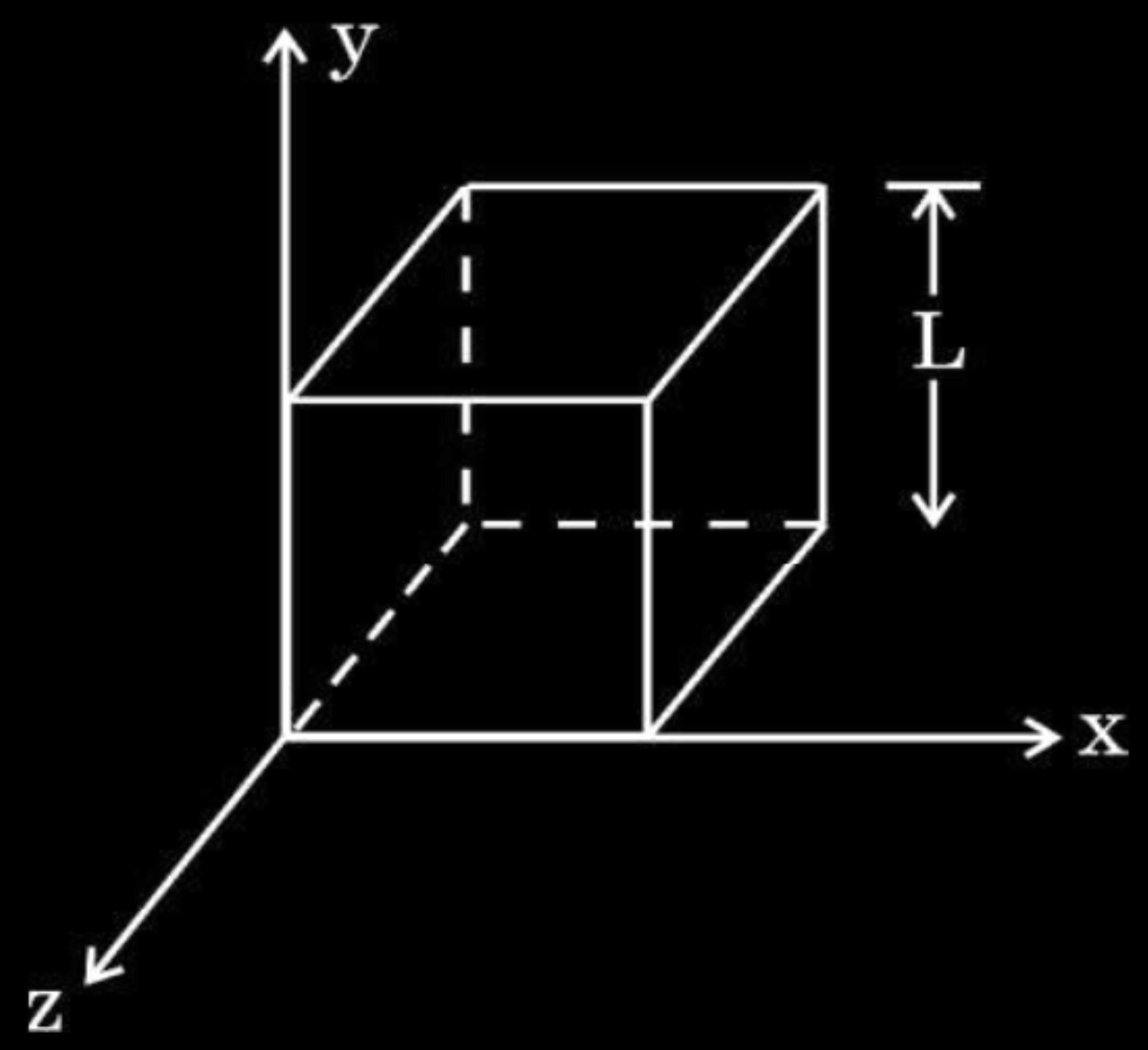
(d) $\dfrac{q\lambda}{\varepsilon_0 r}$

(b) $\dfrac{q\lambda}{2\pi \epsilon_0 r}$

32. (a) (i) Define electric flux and write its SI unit.

(ii) Use Gauss' law to obtain the expression for the electric field due to a uniformly charged infinite plane sheet.

(iii) A cube of side L is kept in space, as shown in the figure. An electric field $\vec{E} = (Ax + B)\,\hat{i}\ \frac{N}{C}$ exists in the region. Find the net charge enclosed by the cube. 5



(a)

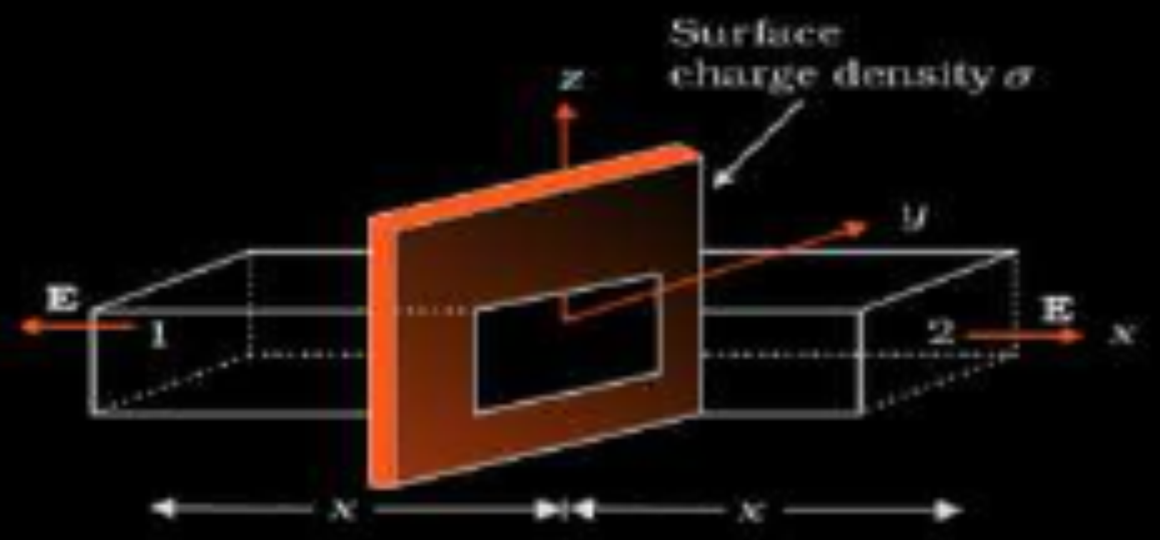
(i) Definition & SI Unit of Electric Flux	½+ ½
(ii) Deriving the expression for electric field due to a uniformly charged infinite plane sheet.	2
(iii) Net charge enclosed by the cube	2

(i) $\phi = \vec{E}.\vec{A}$

Alternatively: Electric flux is the number of electric field lines passing through an area normally. ½

S.I. unit of electric flux Nm^2/C or V-m. ½

(ii)



½

From Gauss's law:- $\phi = \oint \vec{E} \cdot \vec{dA} = \frac{q}{\varepsilon_o}$ ½

$2EA = \frac{\sigma A}{\varepsilon_o}$ ½

$E = \frac{\sigma}{2\varepsilon_o}$ ½

Alternatively: If the shape of the Gaussian surface is taken cylindrical, full credit to be given.

(iii)

$\phi_L = Eds\cos 180^o = -Eds$ ½

$= -BL^2$

$\phi_R = Eds\cos 0^0 = Eds$ ½

$= (AL + B)L^2 = AL^3 + BL^2$

Net flux $= \phi_L + \phi_R$ ½

$= (AL^3 + BL^2) - BL^2$

Net flux $= AL^3 = \frac{q}{\epsilon_o}$ ½

$=> q = AL^3 \epsilon_o$

4. An infinitely long uniformly charged wire produces an electric field of 18×10^4 NC^{-1} at a distance of 1·0 cm. The linear charge density on the wire is :

(a) $1{\cdot}12 \times 10^{-14}$ Cm^{-1}

(b) $3{\cdot}08 \times 10^{-15}$ Cm^{-1}

(c) $1{\cdot}0 \times 10^{-9}$ Cm^{-1}

(d) $1{\cdot}0 \times 10^{-7}$ Cm^{-1}

4. An infinitely long uniformly charged wire produces an electric field of $18 \times 10^4\ NC^{-1}$ at a distance of 1·0 cm. The linear charge density on the wire is :

(a) $1{\cdot}12 \times 10^{-14}\ Cm^{-1}$

(b) $3{\cdot}08 \times 10^{-15}\ Cm^{-1}$

(c) $1{\cdot}0 \times 10^{-9}\ Cm^{-1}$

(d) $1{\cdot}0 \times 10^{-7}\ Cm^{-1}$

(d) $1.0 \times 10^{-7}\ Cm^{-1}$

32. (a) (i) State Coulomb's law in electrostatics and write it in vector form, for two charges.

(ii) 'Gauss's law is based on the inverse-square dependence on distance contained in the Coulomb's law.' Explain.

(iii) Two charges A (charge q) and B (charge 2q) are located at points (0, 0) and (a, a) respectively. Let $\hat{i}$ and $\hat{j}$ be the unit vectors along x-axis and y-axis respectively. Find the force exerted by A on B, in terms of $\hat{i}$ and $\hat{j}$. *5*

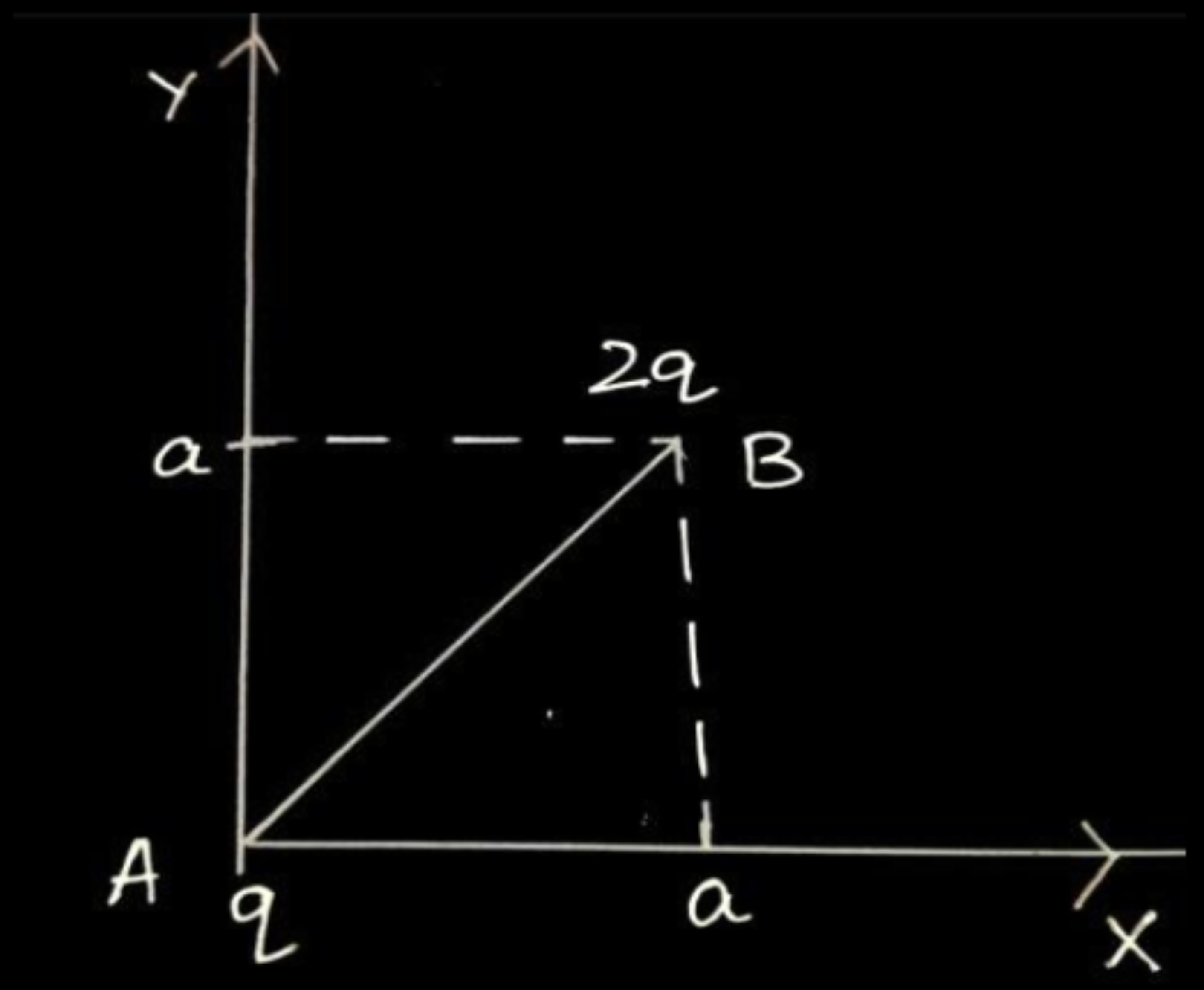
i) Statement of coulomb's law and vector form	1+1
ii) Explanation of Gauss's law based on coulomb's law	1
iii) Force exerted by charge A on charge B	2

i) Force between two point charges varies inversely with the square of distance between the charges and is directly proportional to the product of magnitude of the two charges and acts along the line joining the two charges.

$$\overrightarrow{F_{12}} = \frac{1}{4\pi \in_o} \frac{q_1 q_2}{r_{12}^{\ 2}} \widehat{r_{12}}$$

ii) In derivation of Gauss's law, flux is calculated using Coulomb's law and surface area. Here coulomb's law involves $\frac{1}{r^2}$ factor and surface area involves r^2 factor. When product is taken, the two factors cancel out and flux becomes independent of r.

iii)



$$\vec{r} = \overrightarrow{AB} = a\hat{i} + a\,j$$

1 r = $\left|\overrightarrow{AB}\right| = \sqrt{a^2 + a^2} = \sqrt{2}a$

1 $\vec{F} = \frac{1}{4\pi \in_o} \frac{q_1 q_2}{r^2} \hat{r}$

1 $\vec{F} = \frac{1}{4\pi \in_o} \times \frac{q \times 2q}{(\sqrt{2}a)^2} \times \frac{(a\hat{i} + a\,j)}{\sqrt{2}a}$

$$\vec{F} = \frac{1}{4\pi \in_o} \times \frac{2q^2}{2a^2} \times \frac{(\hat{i} + j)}{\sqrt{2}}$$

$$\vec{F} = \frac{1}{4\pi \in_o} \times \frac{q^2}{\sqrt{2}a^2} \times (\hat{i} + j)$$

½ $\vec{F} = \frac{q^2}{4\sqrt{2}\pi \in_o a^2}(\hat{i} + j)$

Note: Award 1 mark if a student calculates the magnitude of force only.

$$\left|\vec{F}\right| = \frac{1}{4\pi \in_o} \frac{q^2}{a^2}$$

Alternatively

Give full credit if a student uses component method to solve the question.

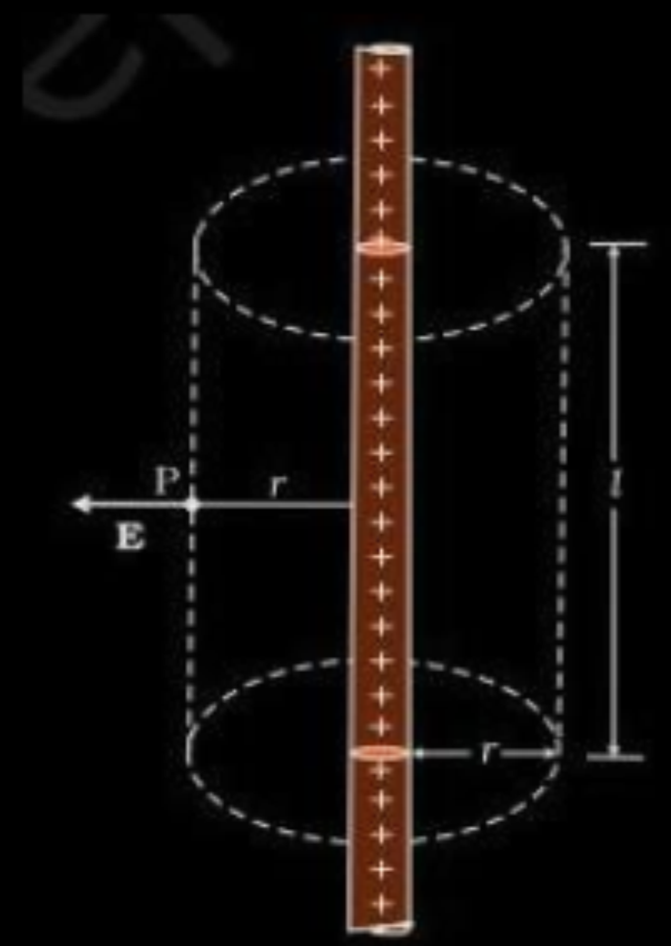
33. (a) (i) Use Gauss' law to obtain an expression for the electric field due to an infinitely long thin straight wire with uniform linear charge density λ.

(ii) An infinitely long positively charged straight wire has a linear charge density λ. An electron is revolving in a circle with a constant speed v such that the wire passes through the centre, and is perpendicular to the plane, of the circle. Find the kinetic energy of the electron in terms of magnitudes of its charge and linear charge density λ on the wire.

(iii) Draw a graph of kinetic energy as a function of linear charge density λ. **5**

(i) Derivation of the expression	-	2
(ii) Finding kinetic energy of electron	-	2
(iii)Graph	-	1

(i)



½

Flux through the Gaussian surface

$\emptyset = E.2\pi rl$

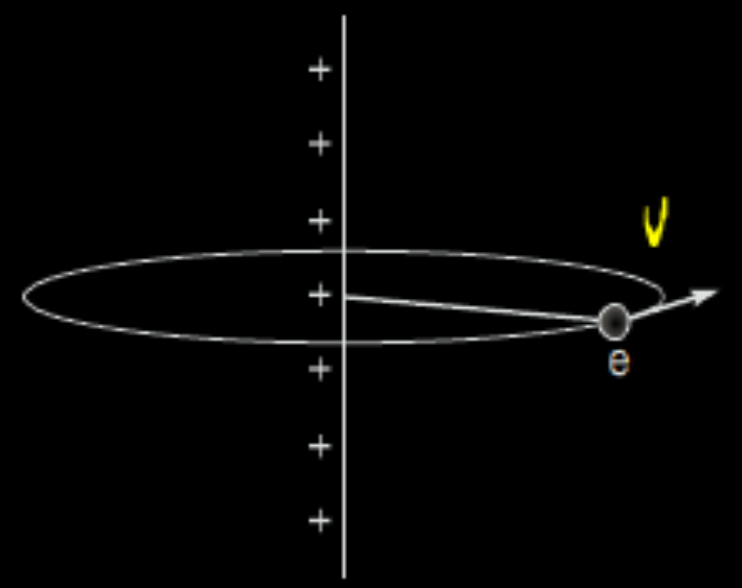
According to Gauss's law

$$E.2\pi rl = \frac{q}{\epsilon_0}$$ ½

$$\therefore \quad q = \lambda l$$ ½

$$E = \frac{\lambda}{2\pi \epsilon_0 r}$$

(i) $$E = \frac{\lambda}{2\pi \epsilon_0 r}$$ ½



½

$$\frac{mv^2}{r} = eE$$

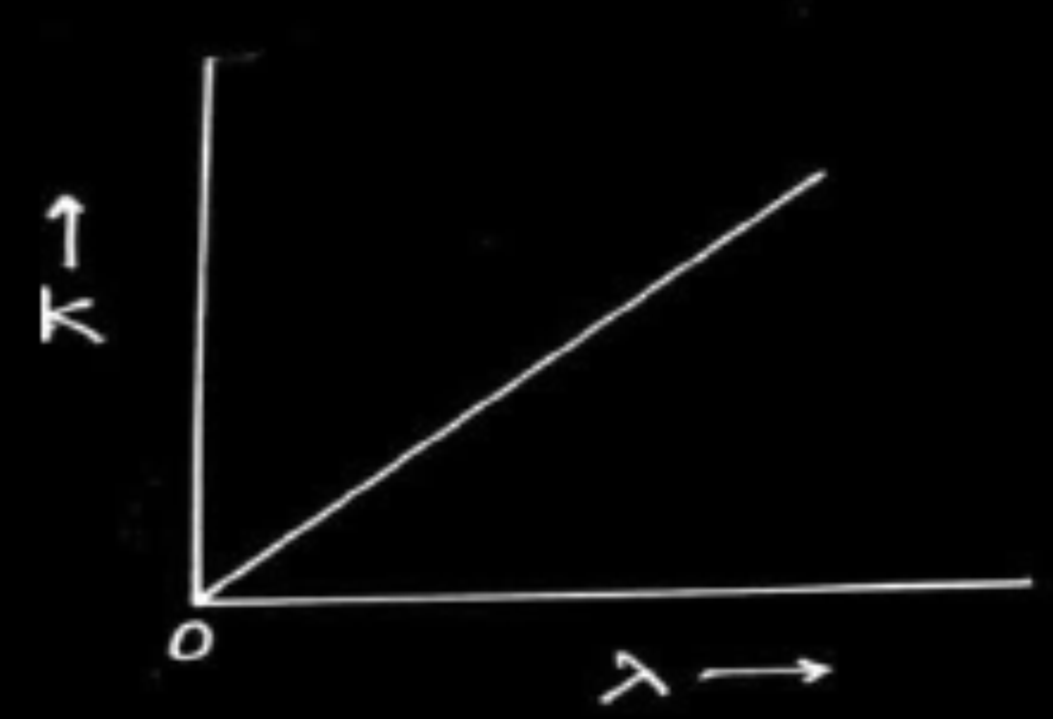
$\therefore$ Kinetic energy $K = \frac{1}{2}mv^2$ ½

$$= \frac{1}{2}eEr$$

$$= \frac{1}{2}e\frac{\lambda.r}{2\pi \epsilon_0 r} = \frac{e\lambda}{4\pi \epsilon_0}$$ ½

(ii) Kinetic energy $K = = \frac{e\lambda}{4\pi \epsilon_0}$ 1

$\therefore \quad K \propto \lambda$



1. A charge Q is placed at the centre of a cube. The electric flux through one if its face is **1**

(A) $\frac{Q}{\varepsilon_0}$ (B) $\frac{Q}{6\varepsilon_0}$

(C) $\frac{Q}{8\varepsilon_0}$ (D) $\frac{Q}{3\varepsilon_0}$

1. A charge Q is placed at the centre of a cube. The electric flux through one if its face is 1

(A) $\frac{Q}{\varepsilon_0}$

(B) $\frac{Q}{6\varepsilon_0}$

(C) $\frac{Q}{8\varepsilon_0}$

(D) $\frac{Q}{3\varepsilon_0}$

(B) $\frac{Q}{6 \in_o}$

1. The electric flux through a Gaussian spherical surface enclosing a point charge q is ϕ. If the charge is replaced by an electric dipole, magnitude of its dipole moment being 2qa, the flux through the surface will be :

(a) $2\,\phi$ (b) ϕ

(c) $\frac{\phi}{2}$ (d) Zero

1. The electric flux through a Gaussian spherical surface enclosing a point charge q is ϕ. If the charge is replaced by an electric dipole, magnitude of its dipole moment being 2qa, the flux through the surface will be :

(a) 2ϕ (b) ϕ

(c) $\frac{\phi}{2}$ (d) Zero

(d) Zero

22. A uniform electric field is represented as $\vec{E} = (3 \times 10^3 \frac{N}{C})\ \hat{i}$. Find the electric flux of this field through a square of side 10 cm when the :

(a) plane of the square is parallel to y-z plane, and

(b) the normal to plane of the square makes an angle of 60° with the x-axis. *2*

(a) Calculation of electric flux	½ + ½
(b) Calculation of electric flux	½ + ½

(a)

$$\Phi = \vec{E}.A\hat{i} = EA$$ ½

$$= 3\times10^{3}\times(100\times10^{-4})$$

$$= 30NC^{-1}m^{2}$$ ½

(b)

$$\Phi = EA\cos 60^{o}$$ ½

$$= 15NC^{-1}m^{2}$$ ½

22. The inward and the outward electric flux through a Gaussian surface are 2ϕ and ϕ respectively.

(a) What is the net charge enclosed by the surface ?

(b) If the net outward flux through the surface were zero, can it be concluded that there were no charges inside the surface ? Justify your answer. *2*

(a)	Calculation of net charge	1
(b)	Conclusion	½
	Justification	½

(a) Net outward flux $= -2\,\Phi + \Phi$	
$= -\Phi$	½
Charge enclosed $= -\Phi\,\varepsilon_o$	½
(b) No	½
There may be charges (positive and negative) and net charge is zero.	½

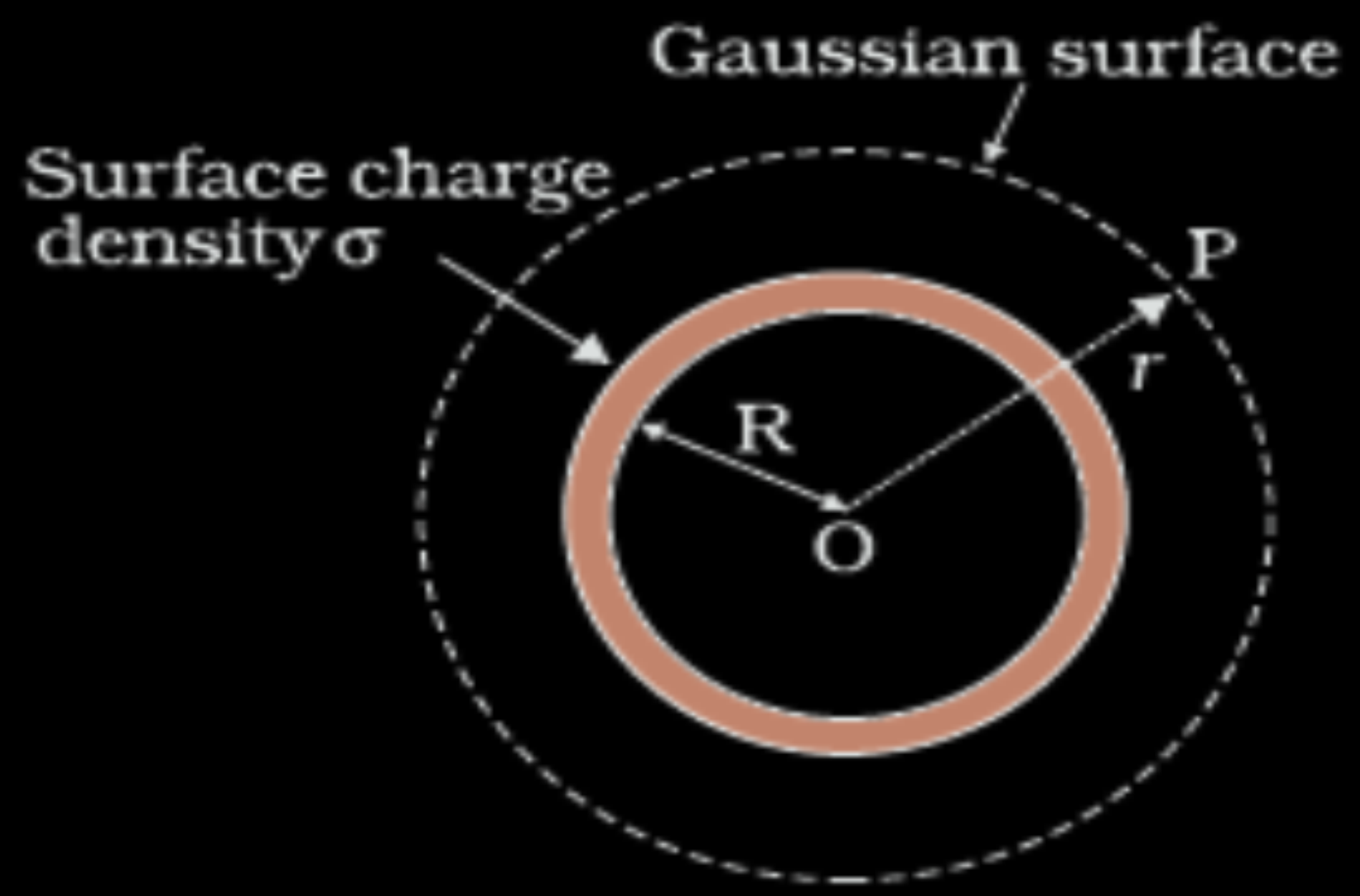
33. (b) (i) A thin spherical shell of radius R has a uniform surface charge density σ. Using Gauss' law, deduce an expression for electric field (i) outside and (ii) inside the shell.

(ii) Two long straight thin wires AB and CD have linear charge densities 10 μC/m and –20 μC/m, respectively. They are kept parallel to each other at a distance 1 m. Find magnitude and direction of the net electric field at a point midway between them.

(i) Deduction of an expression for electric field for (i) and (ii)	3
(ii) Finding magnitude and direction of the net electric field	2

(i)

(i) **Electric Field outside the shell**



½

Electric flux through Gaussian surface

$\Phi = E \times 4\pi r^2$

Charge enclosed by the Gaussian surface

$Q = \sigma \times 4\pi R^2$

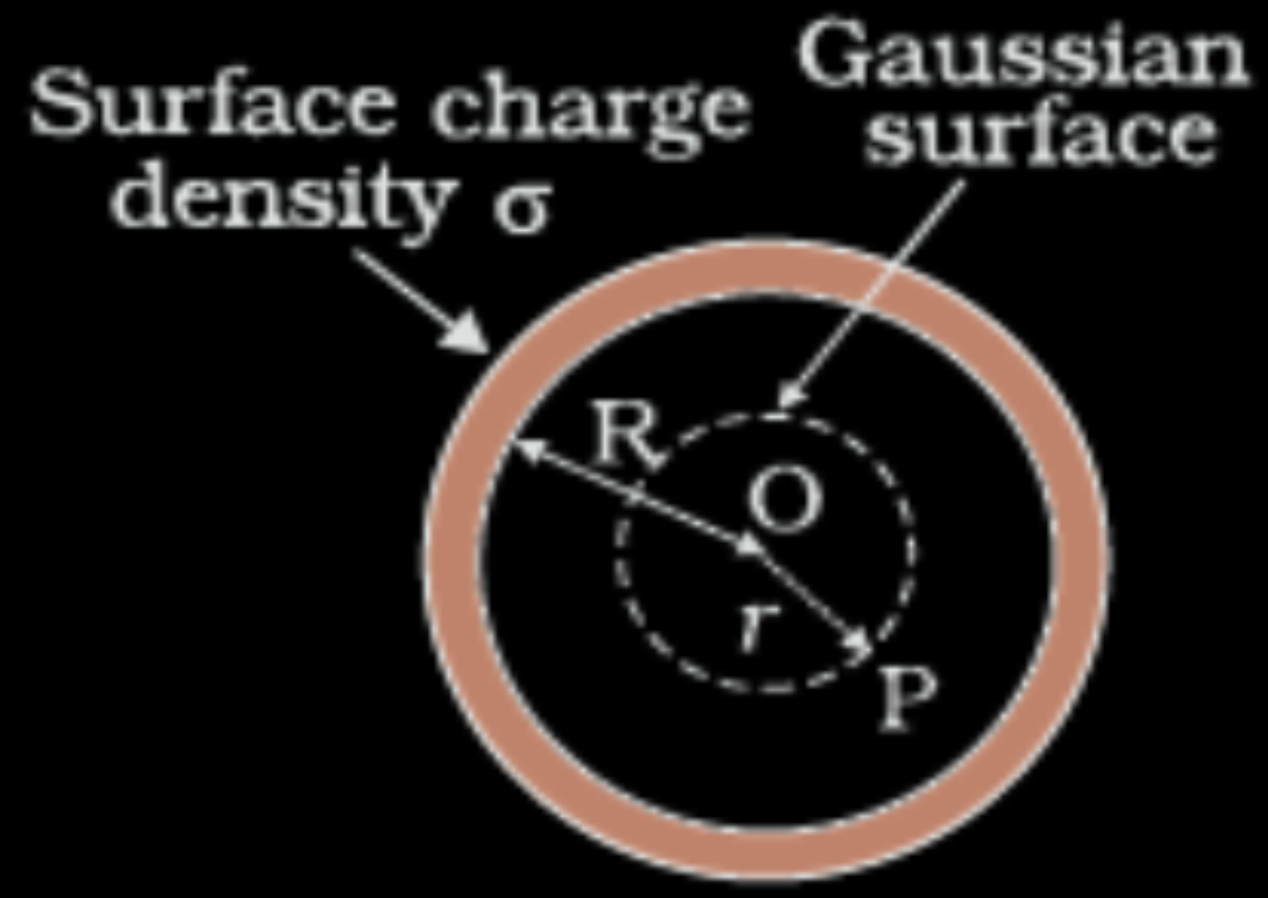
Using Gauss' law: $\int \vec{E}.\vec{ds} = \dfrac{Q}{\varepsilon_0}$ ½

$E \times 4\pi r^2 = \dfrac{(\sigma 4\pi R^2)}{\varepsilon_0}$

$\therefore\ E = \dfrac{\sigma R^2}{\varepsilon_0 r^2}$ ½

$\vec{E} = \dfrac{\sigma R^2}{\varepsilon_0 r^2}\hat{r}$

(ii) **Field inside the shell**



½

Electric flux through Gaussian surface

$\Phi = E \times 4\pi r^2 \qquad (\because r < R)$

Charge enclosed by the Gaussian surface

$Q = 0$

By Gauss' Law

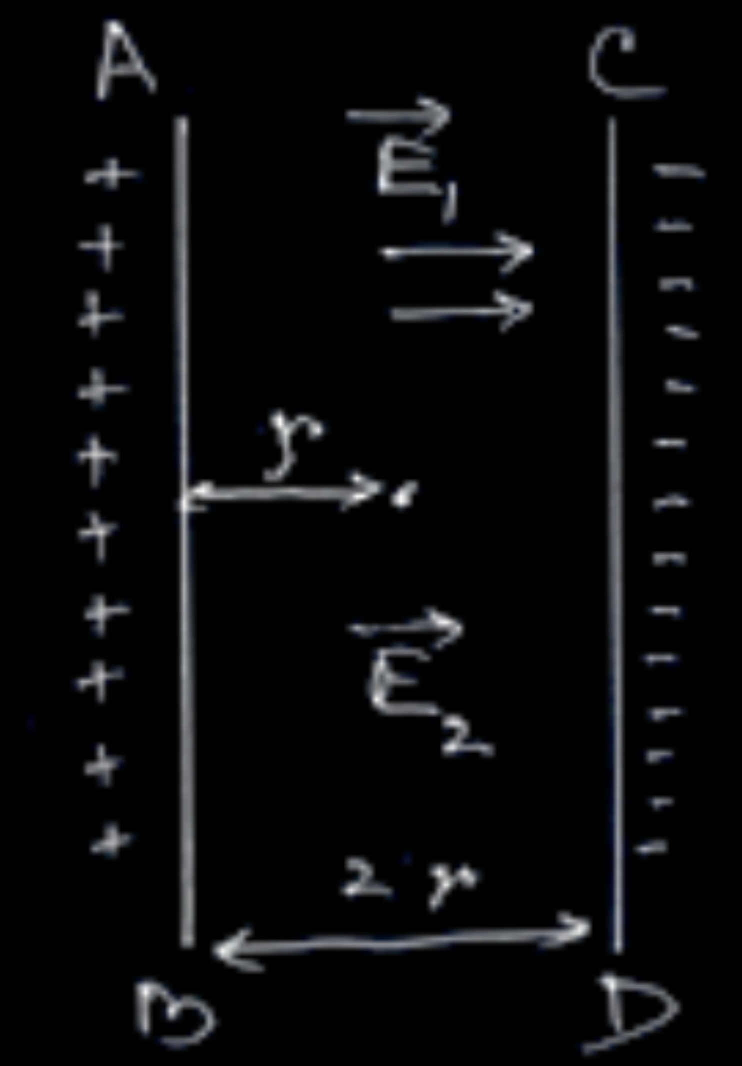
$E \times 4\pi r^2 = 0$ ½

i.e. $E = 0$ ½

(Note: Award full credit of this part if a student writes directly E=0, mentioning as there is no charge enclosed by Gaussian surface)

(ii) Electric field due to a long straight charged wire of linear charged density λ

$$E = \frac{\lambda}{2\pi\varepsilon_0 r}$$ ½



Net electric field at the mid-point

$E_{net} = E_1 + E_2$

$$= \frac{\lambda_1}{2\pi\varepsilon_0 r} + \frac{\lambda_2}{2\pi\varepsilon_0 r}$$ ½

$$E_{net} = \frac{1}{2\pi\varepsilon_0 r}[\lambda_1 + \lambda_2]$$

$$= \frac{2\times 9\times 10^9}{0.5}[10+20]\times 10^{-6}$$

$$= 1.08 \times 10^6 \text{ NC}^{-1}$$ ½

$\vec{E}_{net}$ is directed towards CD. ½

23. (a) Define the term 'electric flux' and write its dimensions.

(b) A plane surface, in shape of a square of side 1 cm is placed in an electric field $\vec{E} = \left(100 \frac{N}{C}\right)\hat{i}$ such that the unit vector normal to the surface is given by $\hat{n} = 0.8\hat{i} + 0.6\hat{k}$. Find the electric flux through the surface.

(a) Defining the term electric flux	1
Writing dimensions	½
(b) Finding the electric flux	1½

(a) It is the measure of the total number of electric field lines passing through a surface normally. **Alternatively**	1
Surface integral of electric field over a surface. **Alternatively** $\phi_E = \vec{E} \cdot \vec{A}$ $\left[ML^3T^{-3}A^{-1} \right]$	½
(b) $\phi_E = \vec{E}.\vec{A}$	½
$= (100\hat{i}).(10^{-4}\ \hat{n})$	
$= (100\hat{i}).(0.8\hat{i} + 0.6\hat{k})10^{-4}$	½
$= 8 \times 10^{-3}\ \mathrm{Nm^2C^{-1}}$	½

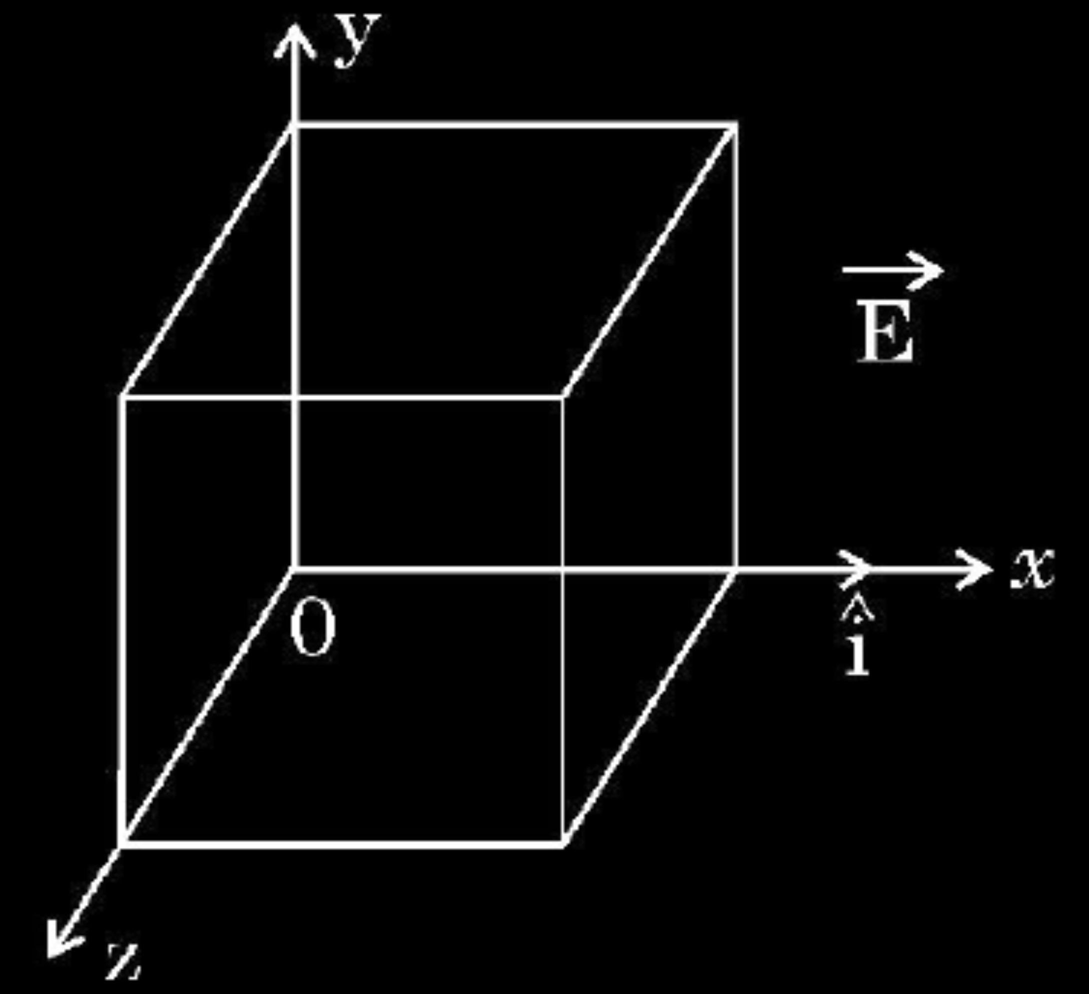
(b) (i) Using Gauss's law, show that the electric field $\vec{E}$ at a point due to a uniformly charged infinite plane sheet is given by $\vec{E} = \frac{\sigma}{2\varepsilon_0}\hat{n}$ where symbols have their usual meanings.

(ii) Electric field $\vec{E}$ in a region is given by

$\vec{E} = (5x^2 + 2)\,\hat{i}$

where E is in N/C and x is in meters.

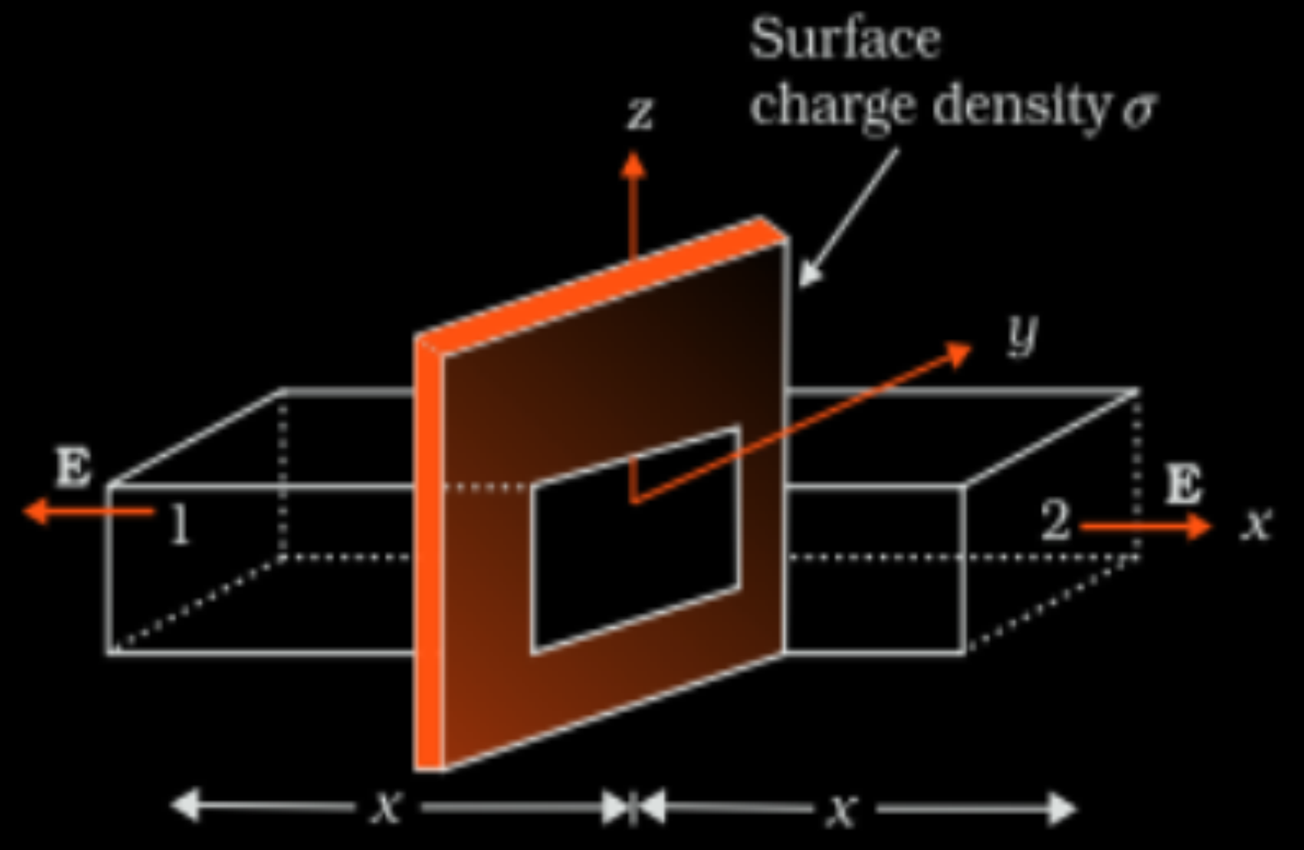
A cube of side 10 cm is placed in the region as shown in figure.



Calculate (1) the electric flux through the cube, and (2) the net charge enclosed by the cube. **5**

i) Showing electric field at a point due to a uniformly charged infinite plane sheet		3
ii) Calculating	(1) electric flux through the cube	1
	(2) charge enclosed by cube	1

(i)



1

$$\oint \vec{E}.d\vec{s} = \int_1 \vec{E}.d\vec{s} + \int_2 \vec{E}.d\vec{s}$$ ½

$$= 2EA$$

From Gauss's law

$$\oint \vec{E}.d\vec{s} = \frac{q}{\varepsilon_0}$$ ½

$$2EA = \frac{\sigma A}{\varepsilon_0}$$

$$E = \frac{\sigma}{2\varepsilon_0}$$ ½

Vectorially $\vec{E} = \frac{\sigma}{2\varepsilon_0}\hat{n}$ ½

Electric field is normally outward of the sheet.

(ii) (1) Electric flux through the cube

$$\phi = \phi_L + \phi_R$$ ½

$$\phi = \int \vec{E}_L.d\vec{s} + \int \vec{E}_R.d\vec{s}$$

$$= -2\times100\times10^{-4} + [5\times(10\times10^{-2})^2 + 2]\times100\times10^{-4}$$

$$\phi = 5\times10^{-4}\ \mathrm{Nm^2C^{-1}}$$ ½

(2)

$$\phi = \frac{q_{en}}{\varepsilon_0}$$ ½

$$q_{en} = \phi.\varepsilon_0$$

$$= 5\times10^{-4}\times8.85\times10^{-12}$$

$$= 4.43\times10^{-15}\ \mathrm{C}$$ ½

2. An infinite long straight wire having a charge density λ is kept along $y'y$ axis in x-y plane. The Coulomb force on a point charge q at a point P $(x, 0)$ will be **1**

(A) attractive and $\dfrac{q\lambda}{2\pi\varepsilon_0 x}$

(B) repulsive and $\dfrac{q\lambda}{2\pi\varepsilon_0 x}$

(C) attractive and $\dfrac{q\lambda}{\pi\varepsilon_0 x}$

(D) repulsive and $\dfrac{q\lambda}{\pi\varepsilon_0 x}$

2. An infinite long straight wire having a charge density λ is kept along y′y axis in x-y plane. The Coulomb force on a point charge q at a point P (x, 0) will be **1**

(A) attractive and $\frac{q\lambda}{2\pi\varepsilon_0 x}$

(B) repulsive and $\frac{q\lambda}{2\pi\varepsilon_0 x}$

(C) attractive and $\frac{q\lambda}{\pi\varepsilon_0 x}$

(D) repulsive and $\frac{q\lambda}{\pi\varepsilon_0 x}$

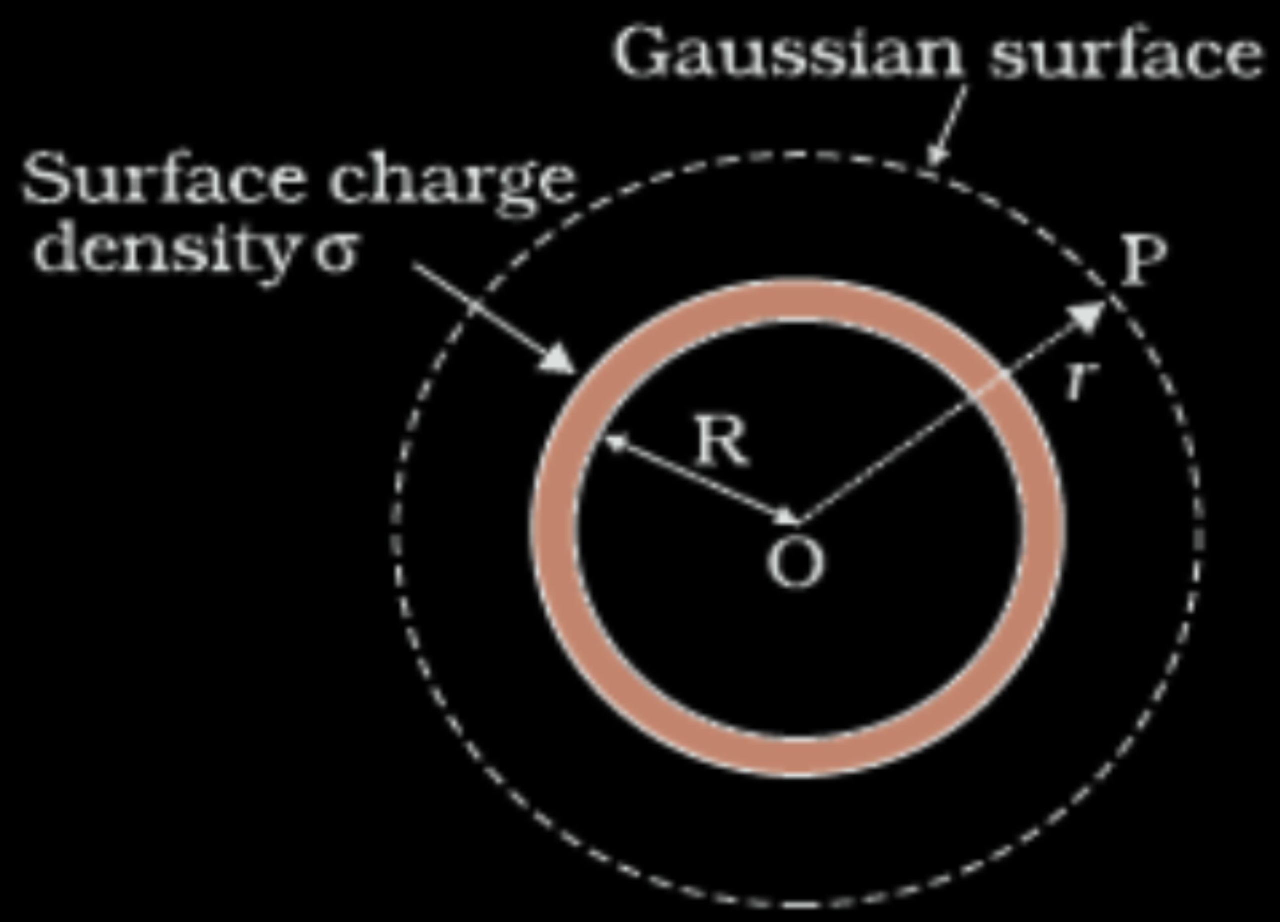
(B) Repulsive and $\frac{q\lambda}{2\pi\varepsilon_0 x}$

(b) (i) A charge + Q is placed on a thin conducting spherical shell of radius R. Use Gauss's theorem to derive an expression for the electric field at a point lying (i) inside and (ii) outside the shell.

(ii) Show that the electric field for same charge density (σ) is twice in case of a conducting plate or surface than in a nonconducting sheet. *5*

(i) Expression for electric field at appoint lying	
(i) inside	1
(ii) outside	2
(ii) Explanation	2

(i) Field inside the shell



The Flux through the Gaussian surface is

$= E \times 4\pi R^2$ ½

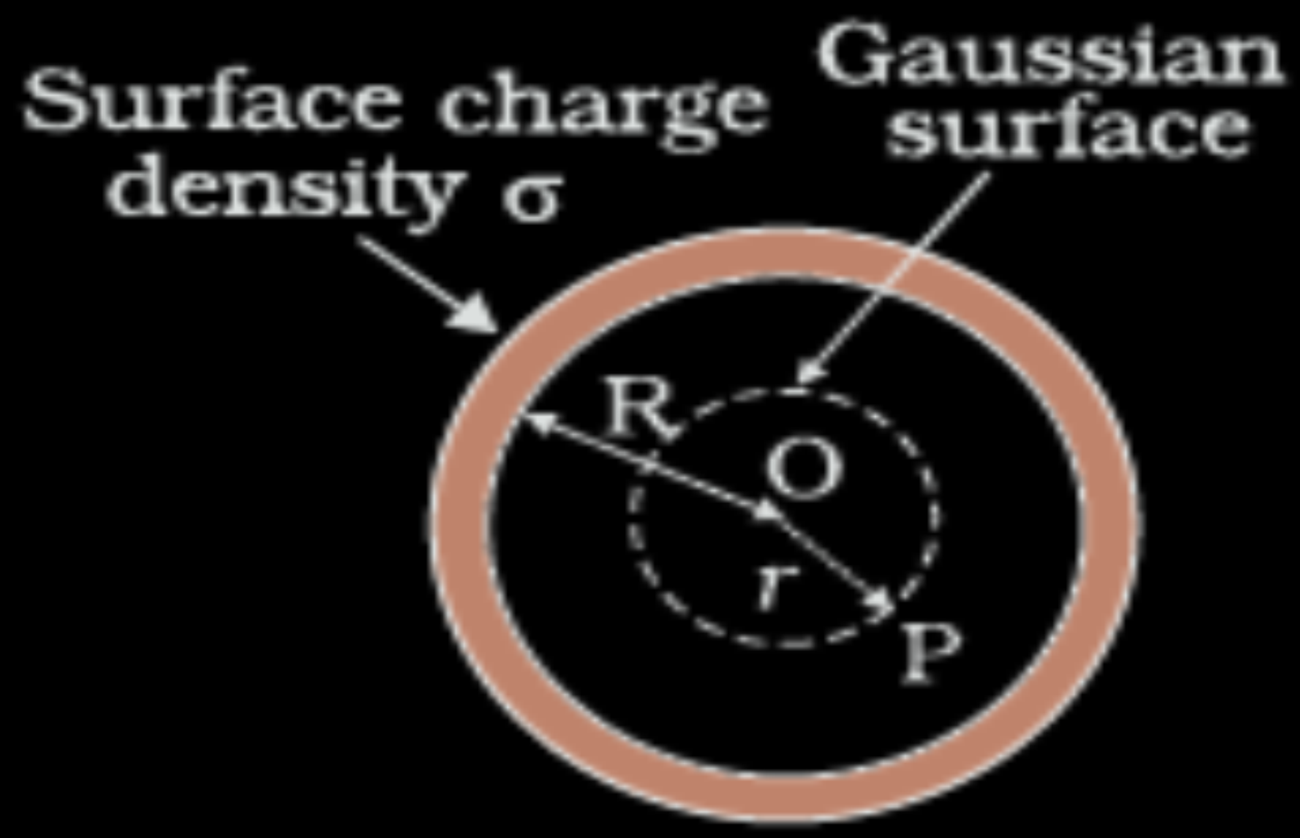
In this case Gaussian surface enclosed no charge.

Hence $E \times 4\pi R^2 = 0$

$E = 0$ ½

(Note: Award full credit of this part if a student writes directly E=0, mentioning as there is no charge enclosed by Gaussian surface)

(ii) Field outside the shell-



½

Electric flux through Gaussian surface

$$E \times 4\pi r^2 = \frac{(\sigma\, 4\pi R^2)}{\varepsilon_0}$$ ½

Charge enclosed by the Gaussian surface

$$E \times 4\pi r^2 = \frac{(\sigma\, 4\pi R^2)}{\varepsilon_0}$$

Using Gauss's law:

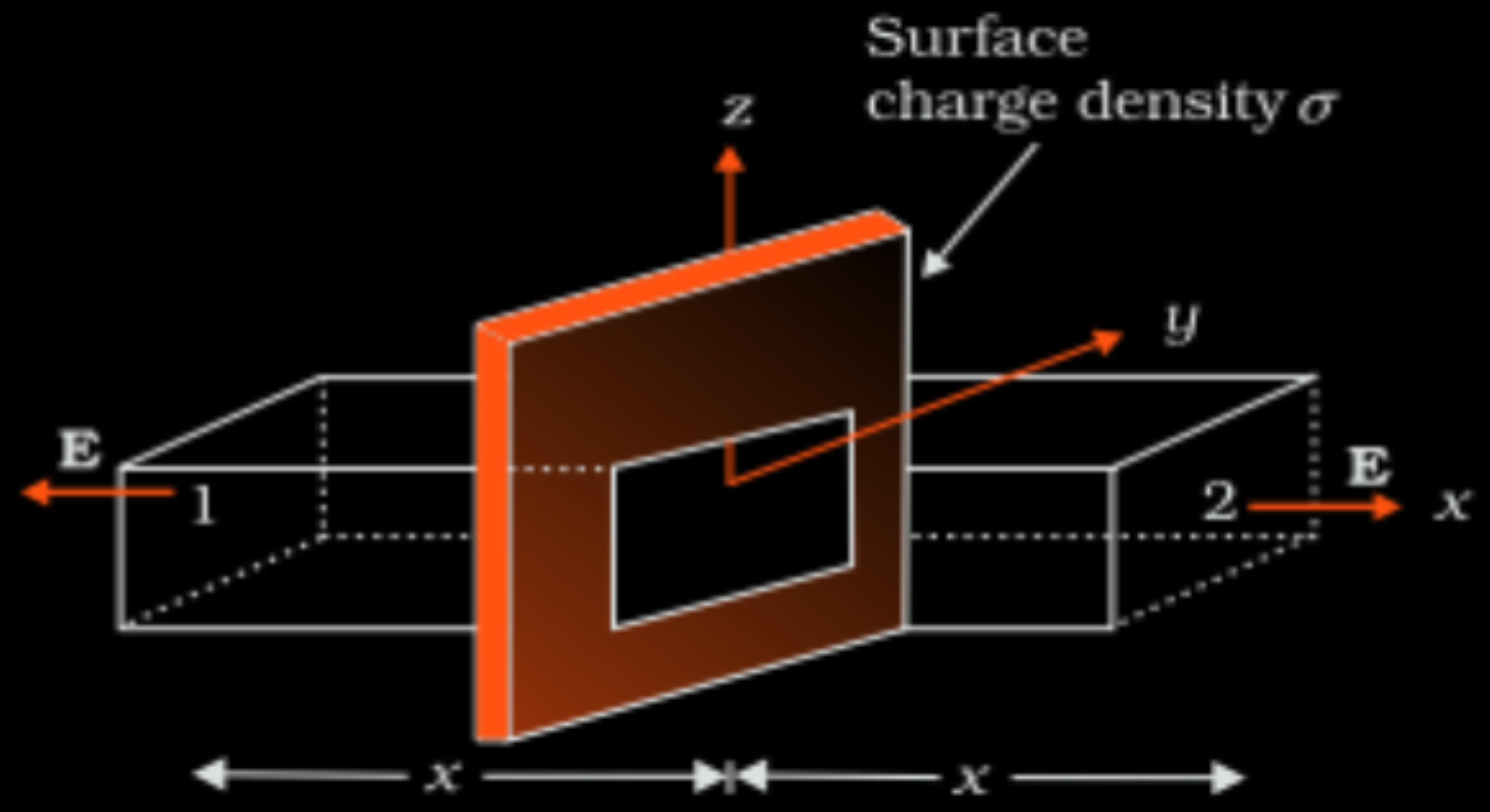
$$\int \vec{E}.\overrightarrow{ds} = \frac{Q}{\varepsilon_0}$$ ½

$$E \times 4\pi r^2 = \frac{(\sigma\, 4\pi R^2)}{\varepsilon_0}$$

$$E = \frac{\sigma}{\varepsilon_0}\frac{R^2}{r^2} = \frac{q}{4\pi\varepsilon_0 r^2}$$ ½

(ii) For conducting sheet,
Electric field due to a conducting sheet

$$E_c = \frac{\sigma}{\varepsilon_0}$$ ½



For non-conducting sheet

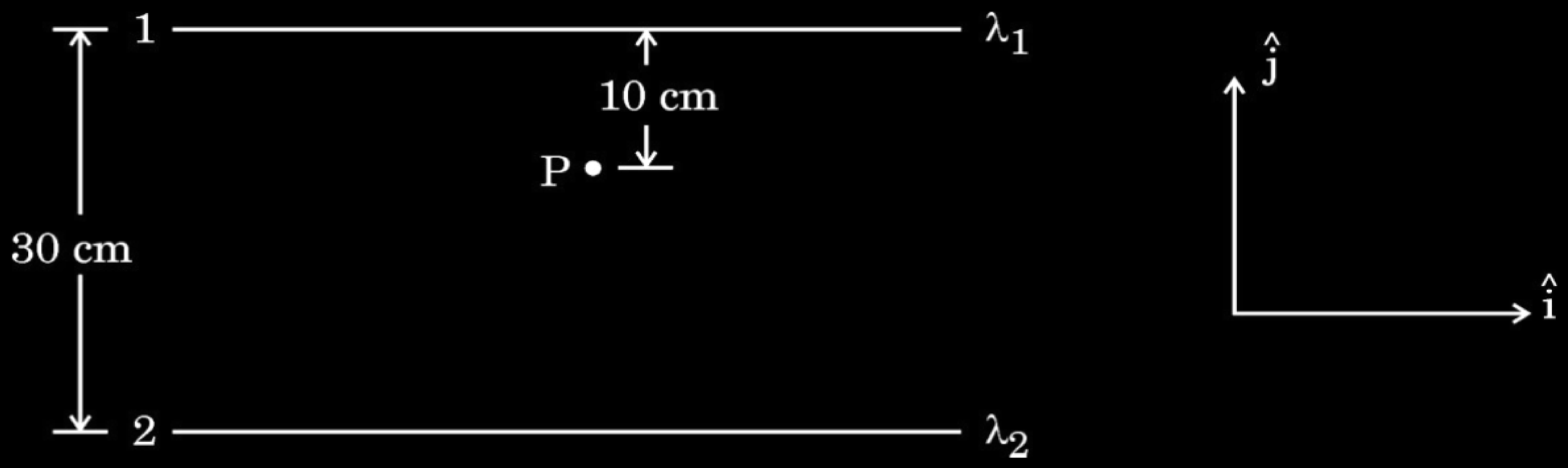
$$E_{nc} = \frac{\sigma}{2\varepsilon_0}$$ ½

Since surface charge density is same. ½

$$2E_{nc} = E_c$$ ½

(b) (i) State Gauss's Law in electrostatics. Apply this to obtain the electric field $\vec{E}$ at a point near a uniformly charged infinite plane sheet.

(ii) Two long straight wires 1 and 2 are kept as shown in the figure. The linear charge density of the two wires are $\lambda_1 = 10\ \mu C/m$ and $\lambda_2 = -20\ \mu C/m$. Find the net force $\vec{F}$ experienced by an electron held at point P. *5*



(i) Statement of Gauss's Law	1
Obtaining expression for electric field	2
(ii) Finding net force on electron	2

1. A thin plastic rod is bent into a circular ring of radius R. It is uniformly charged with charge density λ. The magnitude of the electric field at its centre is :

(A) $\frac{\lambda}{2\varepsilon_0 R}$ (B) Zero (C) $\frac{\lambda}{4\pi\varepsilon_0 R}$ (D) $\frac{\lambda}{4\varepsilon_0 R}$

1. A thin plastic rod is bent into a circular ring of radius R. It is uniformly charged with charge density λ. The magnitude of the electric field at its centre is :

(A) $\frac{\lambda}{2\varepsilon_0 R}$ (B) Zero (C) $\frac{\lambda}{4\pi\varepsilon_0 R}$ (D) $\frac{\lambda}{4\varepsilon_0 R}$

(B) Zero

2. A charged sphere of radius r has surface charge density σ. The electric field on its surface is E. If the radius of the sphere is doubled, keeping charge density the same, the ratio of the electric field on the old sphere to that on the new sphere will be :

(A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) 4

2. A charged sphere of radius r has surface charge density σ. The electric field on its surface is E. If the radius of the sphere is doubled, keeping charge density the same, the ratio of the electric field on the old sphere to that on the new sphere will be :

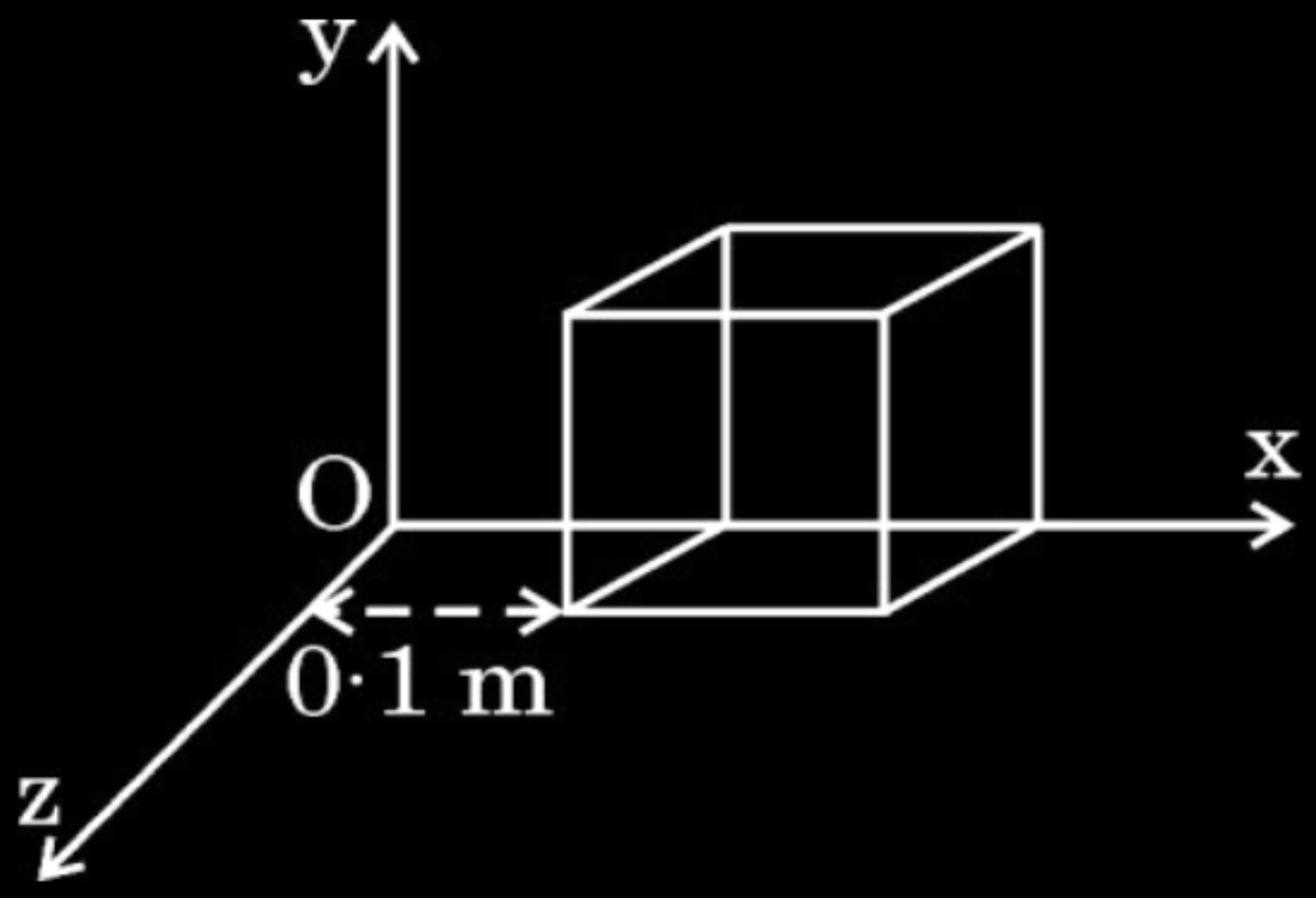
(A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) 4

(A) 1

22. A cube of side 0·1 m is placed, as shown in the figure, in a region where electric field $\vec{E} = 500\, x\, \hat{i}$ exists. Here x is in meters and E in NC^{-1}. Calculate : *3*

(a) the flux passing through the cube, and

(b) the charge within the cube.



Calculating	
(a) the flux passing through the cube	2
(b) the charge within the cube	1

a) $\varphi_L = \overrightarrow{E_L} \cdot \vec{A} = -[500 \times 0.1] \times [(0.1)^2] = -0.5\ \text{N m}^2\ \text{C}^{-1}$ ½

$\varphi_R = \overrightarrow{E_R} \cdot \vec{A} = [500 \times 0.2] \times [(0.1)^2] = 1\ \text{N m}^2\ \text{C}^{-1}$ ½

Net flux $= \varphi_L + \varphi_R = 0.5\ \text{N m}^2\ \text{C}^{-1}$ 1

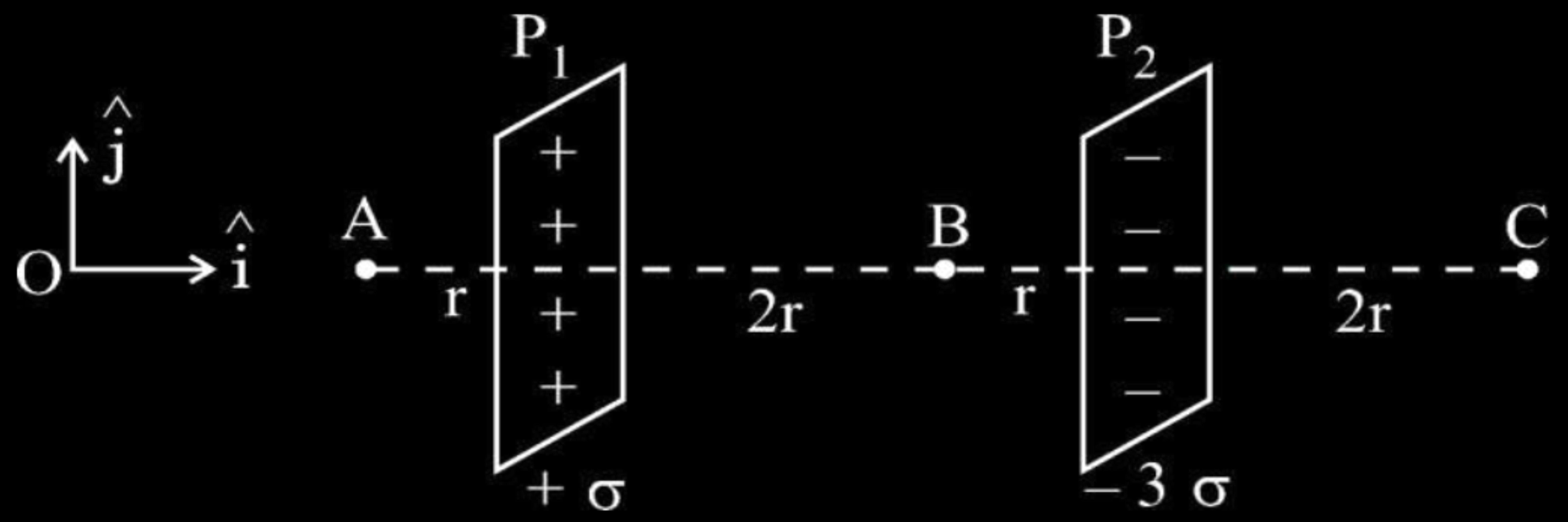
b) flux, $\varphi = \dfrac{q}{\varepsilon_o}$ ½

charge, $q = \varphi \times \varepsilon_o$

$= 0.5\ \varepsilon_o$ ½

$= 4.4 \times 10^{-12}$ C

(b) Two large plane sheets P_1 and P_2 having charge densities $+\sigma$ and -3σ respectively are arranged parallel to each other as shown in the figure. Find the net electric field ($\vec{E}$) at points A, B and C. *3*



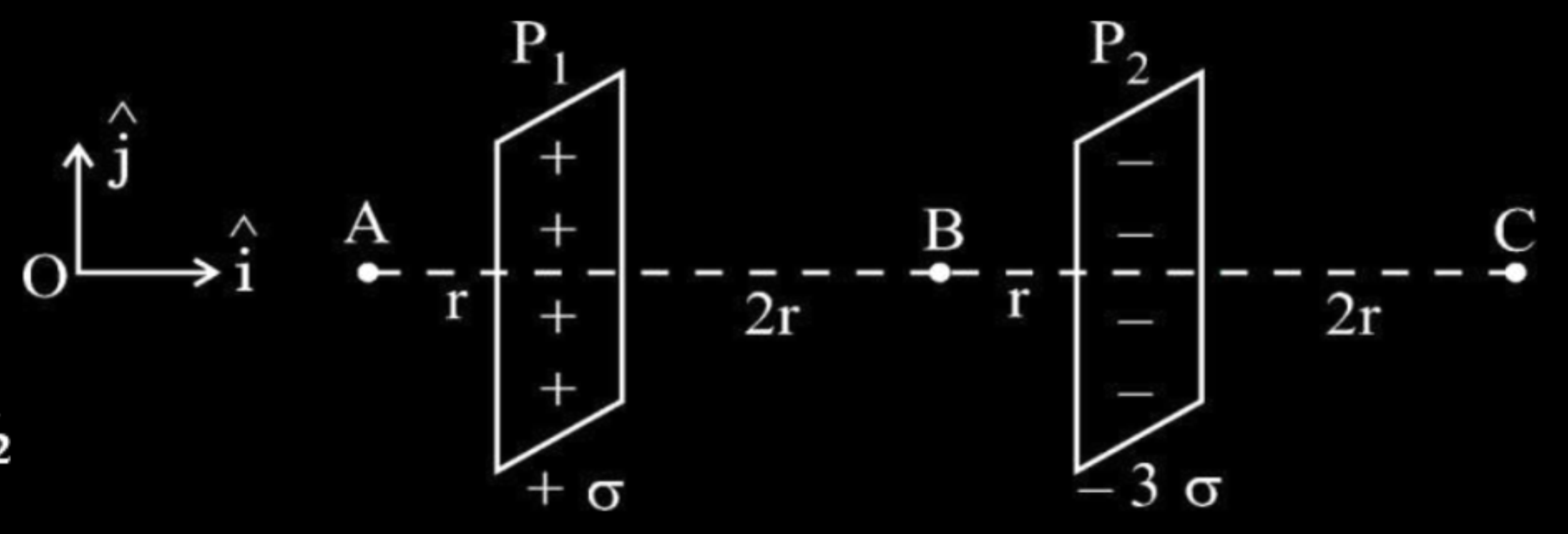
Finding the net electric field (E) at points A,B & C	1+1+1

Electric field at A($\vec{E}_A$)

$$\vec{E}_A = \vec{E}_1 + \vec{E}_2$$

$$= \frac{\sigma}{2\varepsilon_0}(-\hat{i}) + \frac{3\sigma}{2\varepsilon_0}(\hat{i})$$ ½

$$\vec{E}_A = \frac{\sigma}{\varepsilon_0}(\hat{i})$$ ½



Electric field at B ($\vec{E}_B$)

$$\vec{E}_B = \vec{E}_1 + \vec{E}_2$$

$$= \frac{\sigma}{2\varepsilon_0}\hat{i} + \frac{3\sigma}{2\varepsilon_0}\hat{i}$$ ½

$$= \frac{2\sigma}{\varepsilon_0}\hat{i}$$ ½

Electric field at C ($\vec{E}_C$)

$$\vec{E}_c = \vec{E}_1 + \vec{E}_2$$

$$= \frac{\sigma}{2\varepsilon_0}\hat{i} + \frac{3\sigma}{2\varepsilon_0}(-\hat{i})$$ ½

$$= \frac{\sigma}{\varepsilon_0}(-\hat{i})$$ ½

23. An electric field $\vec{E}$ given by :

$$\vec{E} = 100\,\hat{i}\,\frac{N}{C} \qquad \text{for } x > 0$$

$$= -\,100\,\hat{i}\,\frac{N}{C} \qquad \text{for } x < 0$$

exists in a region. A right circular cylinder of length 10 cm and radius 2 cm, is placed in the region such that its axis coincides with x-axis and its two faces are at x = – 5 cm and x = 5 cm. Calculate :

(a) the net outward flux through the cylinder, and

(b) the net charge inside the cylinder. *3*

Calculating	
(a) Net outward flux through the cylinder	2
(b) Net charge inside the cylinder	1

$$\phi_L = \vec{E}.\overrightarrow{\Delta S}$$

$$= 100\left(-\hat{i}\right).\Delta S\left(-\hat{i}\right)$$ ½

$$= 100 \times \pi\left(2\times10^{-2}\right)^2$$

$$= 4\pi\times10^{-2}\ \mathrm{Nm^2/C}$$ ½

$$\phi_R = 100\left(\hat{i}\right).\Delta S\left(\hat{i}\right)$$

$$= 100 \times \pi\left(2\times10^{-2}\right)^2$$

$$= 4\pi\times10^{-2}\ \mathrm{N\,m^2/C}$$

$$\phi_{\mathrm{total}} = \phi_{\mathrm{L}} + \phi_{\mathrm{R}}$$ ½

$$= 8\pi\times10^{-2}\ \mathrm{Nm^2/C}$$ ½

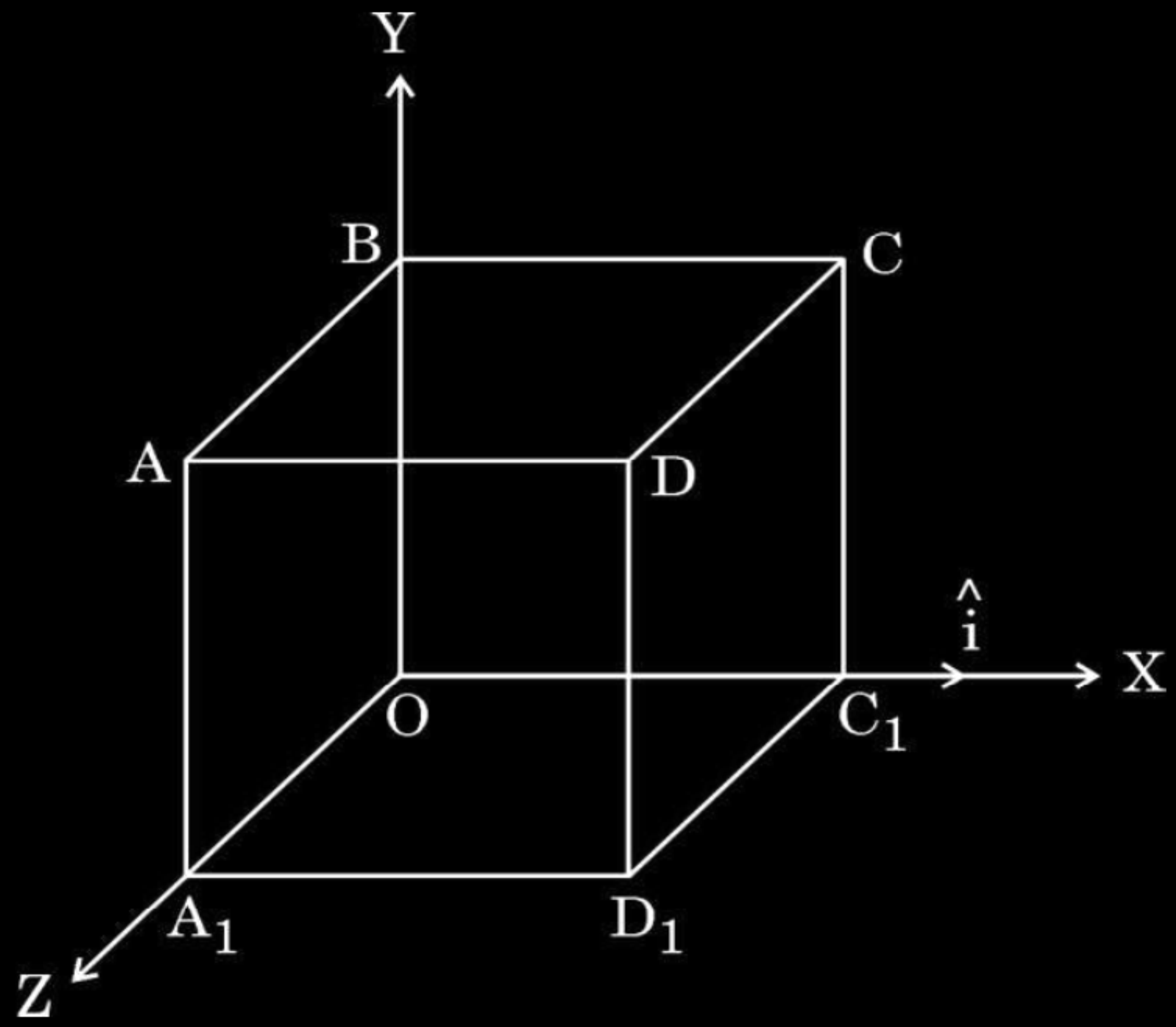
$$= 25.12\times10^{-2}\ \mathrm{Nm^2/C}$$ ½

Charge $q = \varepsilon_0\,\phi_{total}$

$$= 25.12\times10^{-2}\times8.85\times10^{-12}$$ ½

$$= 0.22\times10^{-11}\ \mathrm{C}$$

(ii) An electric field $\vec{E} = (10x + 5)\hat{i}$ N/C exists in a region in which a cube of side L is kept as shown in the figure. Here x and L are in metres. Calculate the net flux through the cube.



(ii) $\vec{E} = (10x+5)\hat{i}$ N/C

$$\phi_L = \int \vec{E}.\vec{ds}$$

$$= -E_L(L^2)$$

$$= -5L^2$$ ½

$$\phi_R = E_R(L^2)$$

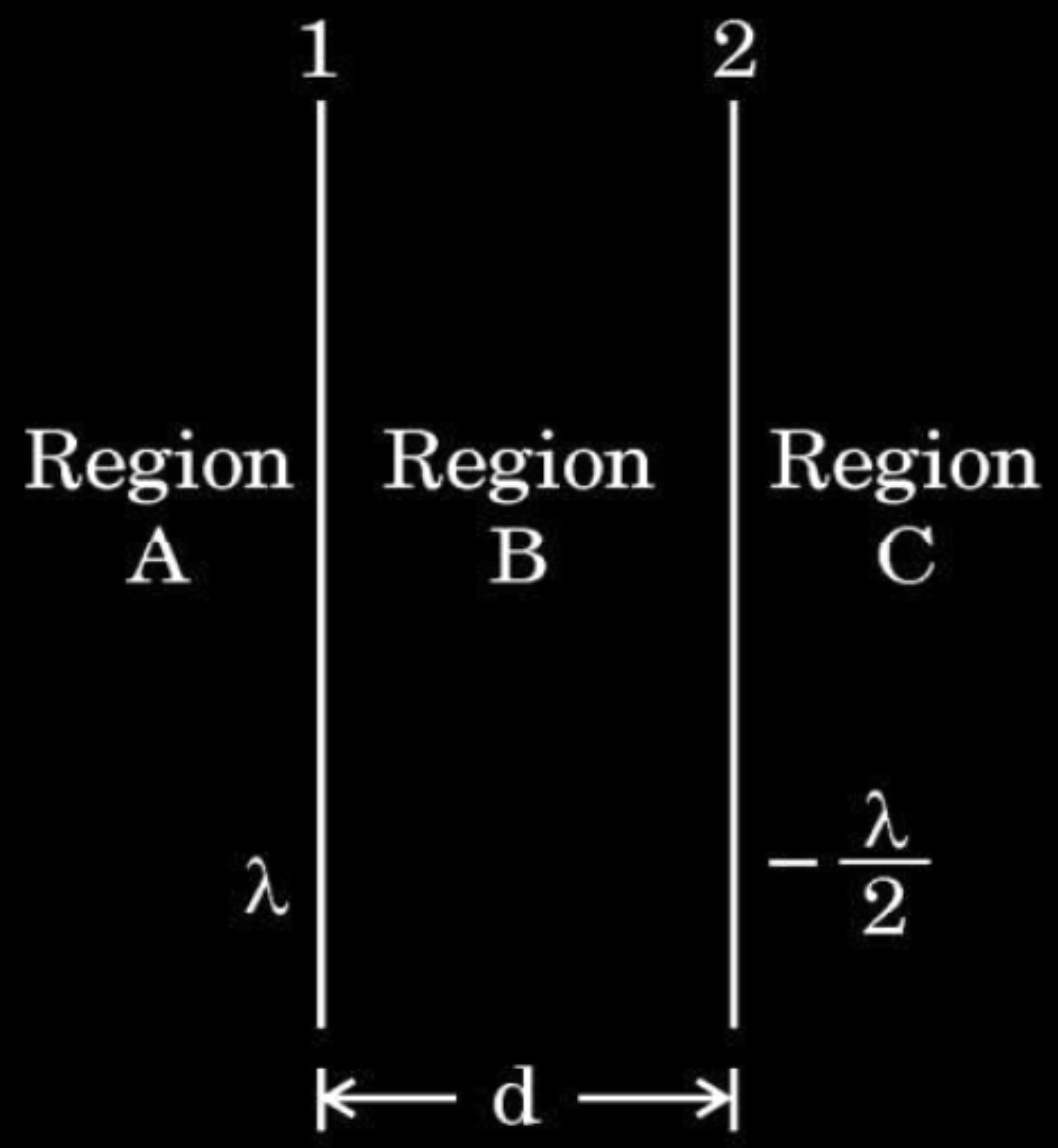
$$= (10L+5)L^2$$ ½

$$\phi_{net} = \phi_L + \phi_R$$ ½

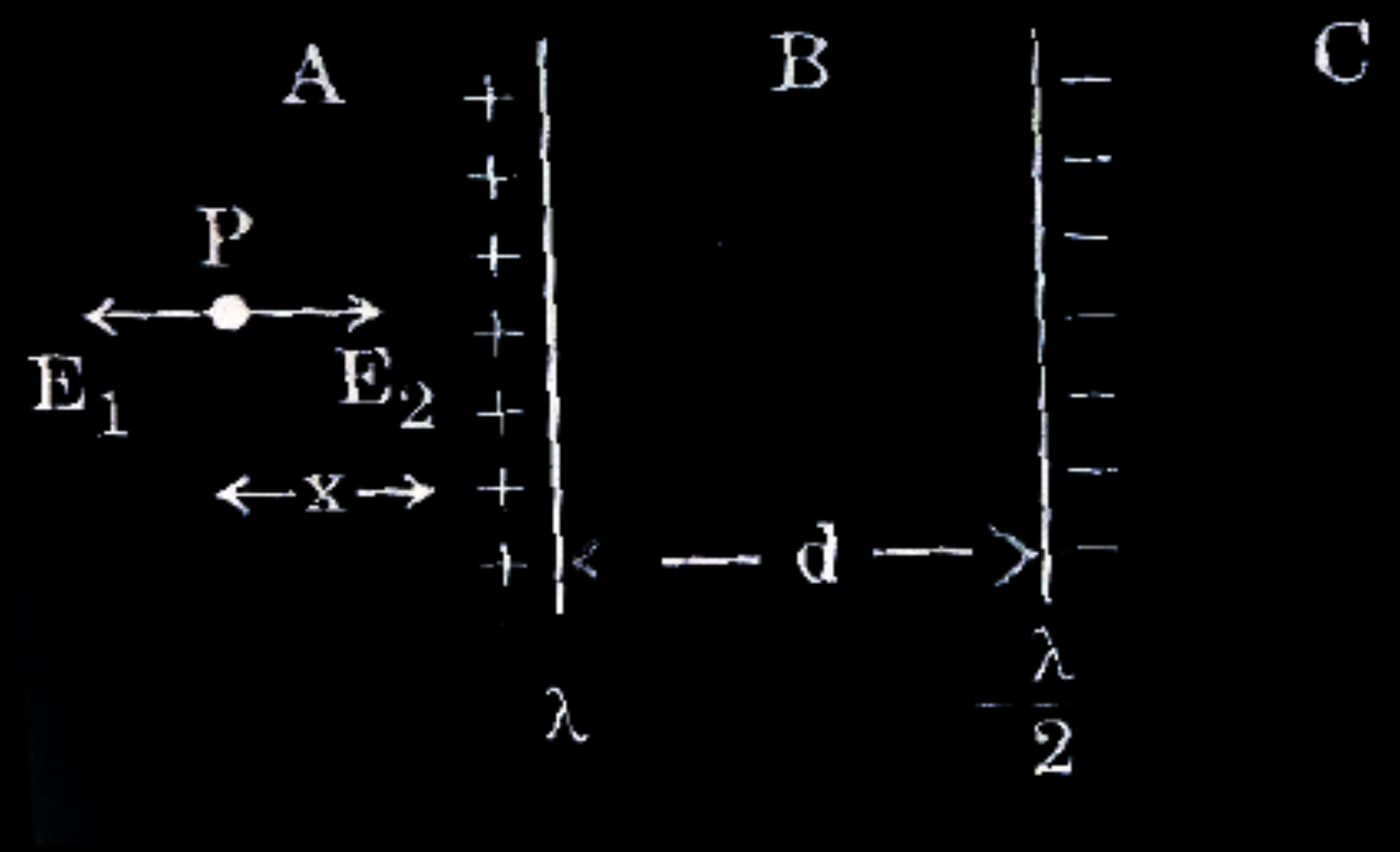
$$= -5L^2 + (10L+5)L^2$$

$$= 10L^3 \ \ \text{Nm}^2/\text{C}$$ ½

(b) Two infinitely long straight wires '1' and '2' are placed d distance apart, parallel to each other, as shown in the figure. They are uniformly charged having charge densities λ and $-\frac{\lambda}{2}$ respectively. Locate the position of the point from wire '1' at which the net electric field is zero and identify the region in which it lies. *3*



Location of point at which net electric field is zero	2½
Identification of Region	½



½

Electric field due to wire 1 and wire 2 at point P

$$E_1 = \frac{\lambda}{2\pi\varepsilon_0 x}$$ ½

$$E_2 = \frac{\lambda/2}{2\pi\varepsilon_0 (x+d)}$$ ½

At P, Net electric field is zero ½

$$E_1 = E_2$$

$$\frac{\lambda}{2\pi\varepsilon_0 x} = \frac{\lambda}{2\times 2\pi\varepsilon_0 (x+d)}$$

$$x = -2d$$ ½

Negative sign indicates that point lies in the region C.

At a distance 2d from wire 1 electric field is zero. ½

(**Note** : Award full credit if a student finds the position by taking point in region C directly)

(b) (i) Show that Gauss's theorem is consistent with Coulomb's law. Using it, derive an expression for the electric field due to a uniformly charged thin spherical shell of radius r at a point at a distance y from the centre of the shell such that (I) $y > r$, and (II) $y < r$.

(i)	
• Showing consistency of Gauss's theorem with Coulomb's law	1
• Derivation for electric field due to uniformly charged thin spherical shell at (I) $y > r$ (II) $y < r$	2

(i)

- Gauss's theorem is based on the inverse square dependence on distance contained in the coulomb's law. 1

Alternatively-

According to Gauss's theorem

$$\oint \vec{E}.d\vec{s} = \frac{q}{\varepsilon_o}$$

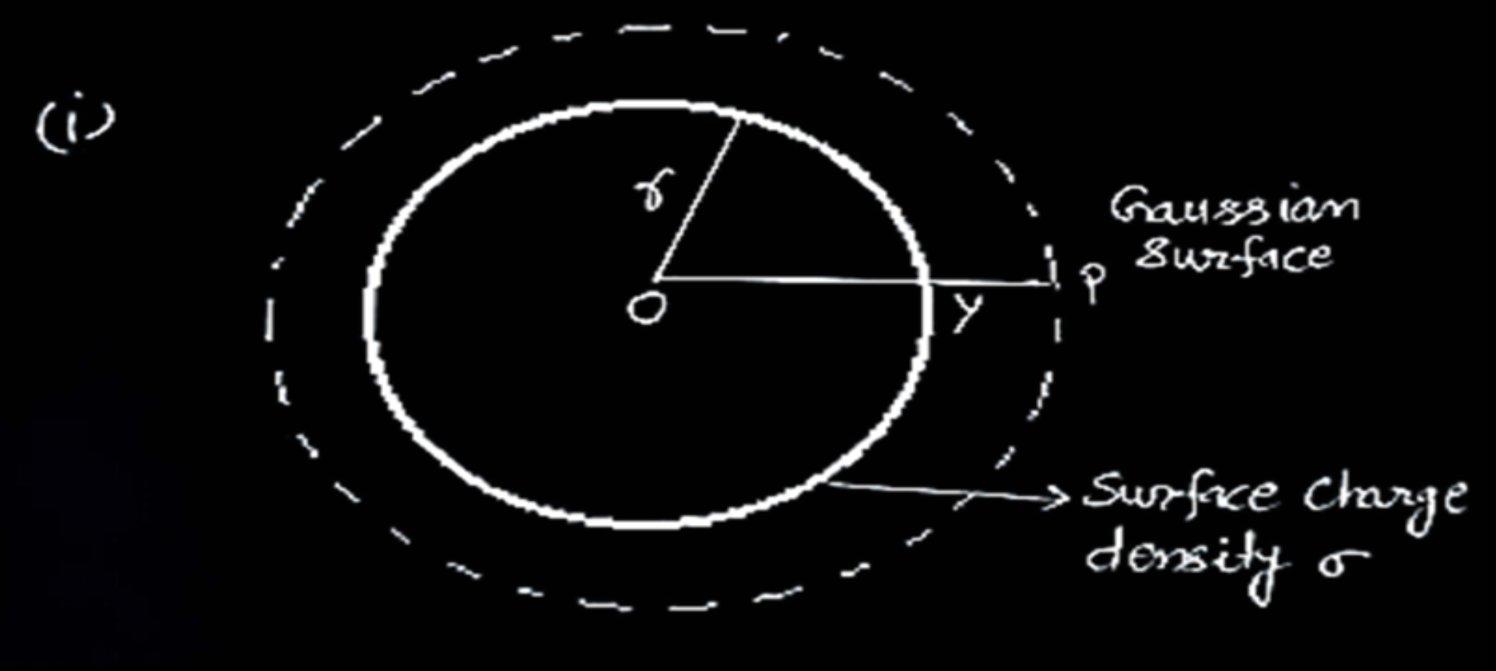
$$E = \frac{1}{4\pi\varepsilon_o}\frac{q}{r^2}$$ ½

According to Coulomb's law, force on charge q_o in this field

$$F = \frac{1}{4\pi\varepsilon_o}\frac{qq_o}{r^2}$$ ½

Therefore, Gauss's law is consistent with Coulomb's law

- (I) For $y > r$



Electric flux through Gaussian surface $E\times 4\pi y^2$

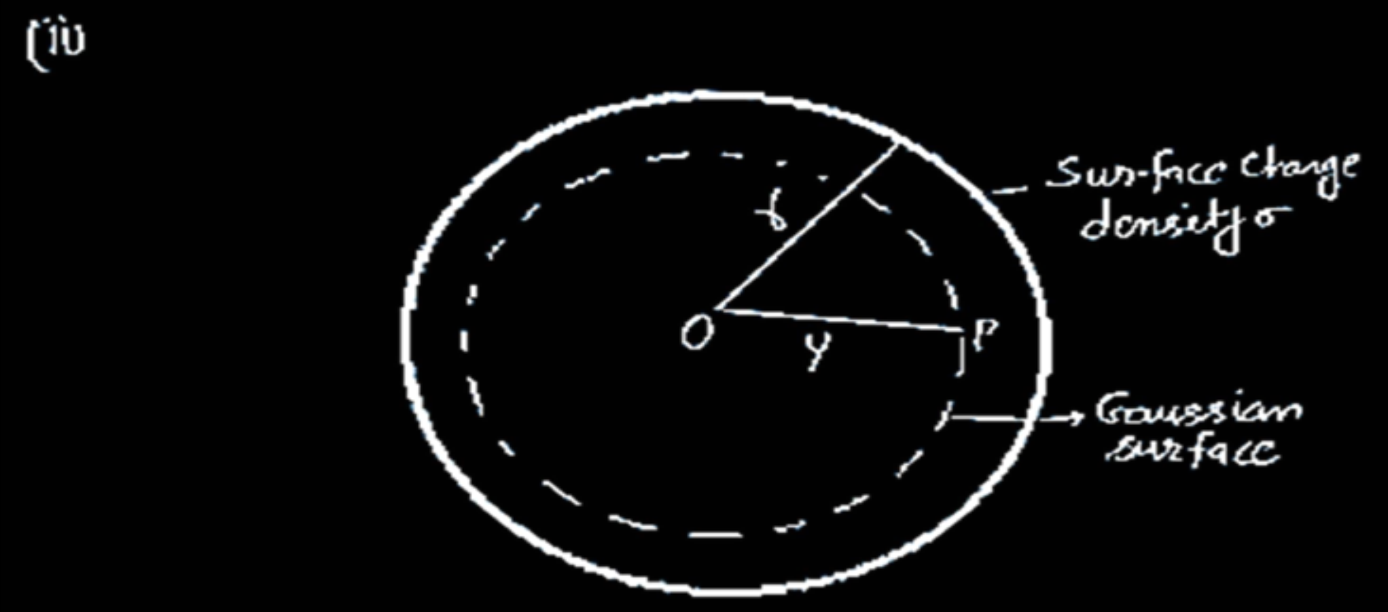
The charge enclosed by the surface $\sigma\times 4\pi r^2$

Using Gauss theorem

$$E(4\pi y^2) = \frac{\sigma 4\pi r^2}{\varepsilon_o}$$ ½

$$\vec{E} = \frac{q}{4\pi\varepsilon_0 y^2}\hat{r}$$ ½

(II) For $y < r$



The charge enclosed by Gaussian surface = 0

Using Gauss theorem

Electric flux $= E(4\pi y^2) = 0$ ½

i.e. $E = 0 \quad (y < r)$ ½

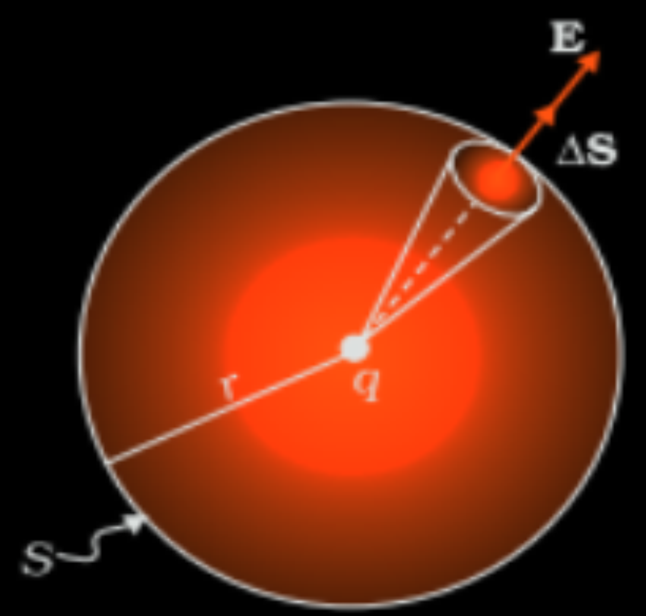
(b) (i) What is difference between an open surface and a closed surface ? Draw elementary surface vector $d\vec{S}$ for a spherical surface S.

(ii) Define electric flux through a surface. Give the significance of a Gaussian surface. A charge outside a Gaussian surface does not contribute to total electric flux through the surface. Why ?

(iii) A small spherical shell S_1 has point charges $q_1 = -3\ \mu C$, $q_2 = -2\ \mu C$ and $q_3 = 9\ \mu C$ inside it. This shell is enclosed by another big spherical shell S_2. A point charge Q is placed in between the two surfaces S_1 and S_2. If the electric flux through the surface S_2 is four times the flux through surface S_1, find charge Q.

(i) Difference between an open surface and a closed surface	½
Diagram of elementary surface vector $\vec{ds}$	1
(ii) Definition of electric flux	1
Significance of Gaussian Surface	½
Reason	½
(iii) Finding charge Q	1½

(i) Open Surface – A surface which does not enclose a volume.
Closed Surface – A surface which does enclose a volume. ½



1

(ii) Electric flux is defined as the number of electric field lines crossing an area normally. 1

Alternatively-

$\phi = \vec{E}.\vec{A}$

Alternatively-

$\phi = EA\cos\theta$

Significance of Gaussian Surface: -

It helps in finding the electric field in a simpler way. ½

Reason: -

Because any electric field line from the charge which enters the surface at one point will exit at another, resulting in a net zero flux. ½

(iii) Total charge enclosed by $S_1 = (-3-2+9)\ \mu C = 4\ \mu C$ ½

Total charge enclosed by $S_2 = Q + 4\ \mu C$

$\phi_{s_2} = 4\phi_{s_1}$ ½

$$\frac{Q + 4\mu C}{\varepsilon_o} = 4\left(\frac{4\mu C}{\varepsilon_o}\right)$$

$Q = 12\ \mu C$ ½

1. Two identical point charges are placed at the two vertices A and B of an equilateral triangle of side l. The magnitude of the electric field at the third vertex P is E. If a hollow conducting sphere of radius $(l/4)$ is placed at P, the magnitude of the electric field at point P now becomes **1**

(A) >E

(B) E

(C) $\frac{E}{2}$

(D) zero

1. Two identical point charges are placed at the two vertices A and B of an equilateral triangle of side l. The magnitude of the electric field at the third vertex P is E. If a hollow conducting sphere of radius ($l/4$) is placed at P, the magnitude of the electric field at point P now becomes **1**

(A) $>E$ (B) E

(C) $\frac{E}{2}$ (D) zero

(D) Zero

(b) (i) Consider three metal spherical shells A, B and C, each of radius R. Each shell is having a concentric metal ball of radius R/10. The spherical shells A, B and C are given charges +6q, –4q, and 14q respectively. Their inner metal balls are also given charges –2q, +8q and –10q respectively. Compare the magnitude of the electric fields due to shells A, B and C at a distance 3R from their centres.

Total charge for A = Total charge for B = Total charge for C = $_{+}4q$ 1

As , $E = \dfrac{kQ}{r^2}$

Since $Q = 4q$ and $r = 3R$ ½

$$E = \frac{k(4q)}{9R^2} = \frac{4kq}{9R^2}$$ ½

$\therefore\ E_A = E_B = E_c$

7. The electric flux through a closed Gaussian surface depends upon

(a) Net charge enclosed and permittivity of the medium

(b) Net charge enclosed, permittivity of the medium and the size of the Gaussian surface

(c) Net charge enclosed only

(d) Permittivity of the medium only **1**

7. The electric flux through a closed Gaussian surface depends upon

(a) Net charge enclosed and permittivity of the medium

(b) Net charge enclosed, permittivity of the medium and the size of the Gaussian surface

(c) Net charge enclosed only

(d) Permittivity of the medium only **1**

(a) Net Charge enclosed and permittivity of the medium

13. A point charge is placed at the centre of a hollow conducting sphere of internal radius ‘r’ and outer radius ‘2r’. The ratio of the surface charge density of the inner surface to that of the outer surface will be ________. **1**

13. A point charge is placed at the centre of a hollow conducting sphere of internal radius 'r' and outer radius '2r'. The ratio of the surface charge density of the inner surface to that of the outer surface will be ________. **1**

$\boxed{4:1}$

(a) State Gauss's law on electrostatics and derive an expression for the electric field due to a long straight thin uniformly charged wire (linear charge density λ) at a point lying at a distance r from the wire.

(a) Statement of Gauss's law	1
Derivation of the expression of the electric field	2½

(a) Gauss law: Electric flux through of a closed surface is $\frac{1}{\varepsilon_0}$ times the charge enclosed by the surface.

Alternatively

$$\phi_E = \frac{q}{\varepsilon_0}$$ 1

Let the charge be uniformly distributed on a wire

$$\phi = \oint d\phi = \int_{s_1} \vec{E} \cdot d\vec{s}_1 + \int_{s_2} \vec{E} \cdot d\vec{s}_2 + \int_{s_3} \vec{E} \cdot d\vec{s}_3$$ ½

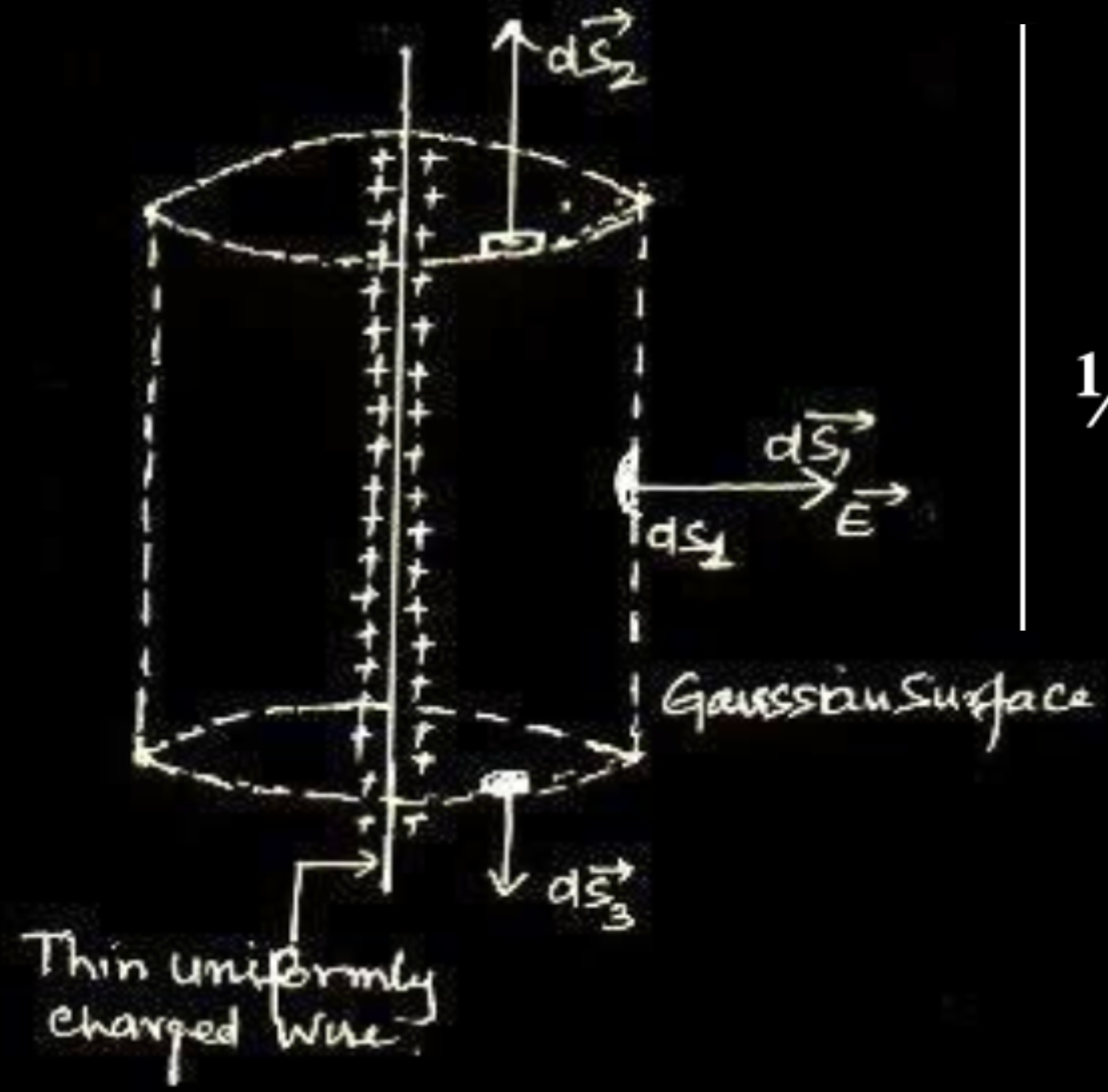
$$= \int_{s_1} Eds_1 \cos 0^0 + \int_{s_2} Eds_2 \cos 90^0 + \int_{s_3} Eds_3 \cos 90^0$$ ½

$$= E\int_{s_1} ds_1 = E.2\pi rl$$

$by Gauss's law$ ½

$$\frac{q}{\varepsilon_0} = E \cdot 2\pi rl$$

$$E = \frac{q}{2\pi\varepsilon_0 rl} = \frac{1}{2\pi\varepsilon_0}\frac{\lambda}{r}$$ ½



½

28. A hollow conducting sphere of inner radius r_1 and outer radius r_2 has a charge Q on its surface. A point charge –q is also placed at the centre of the sphere.

(a) What is the surface charge density on the (i) inner and (ii) outer surface of the sphere ?

(b) Use Gauss' law of electrostatics to obtain the expression for the electric field at a point lying outside the sphere. **3**

(a) Giving the value of surface charge density of (i) Inner surface (ii) Outer Surface	½ + ½
(b) Deriving expression for electric field	2

(a) Surface charge density on the inner surface$= \frac{q}{4\pi r_1^2}$ ½

On the outer surface$= \frac{Q-q}{4\pi r_2^2}$ ½

(b) For a spherical Gaussian surface $x > r_2$

$$\oint \vec{E}\overrightarrow{ds} = \frac{Q-q}{\epsilon_0}$$ 1

$$E \times 4\pi x^2 = \frac{Q-q}{\epsilon_0}$$ ½

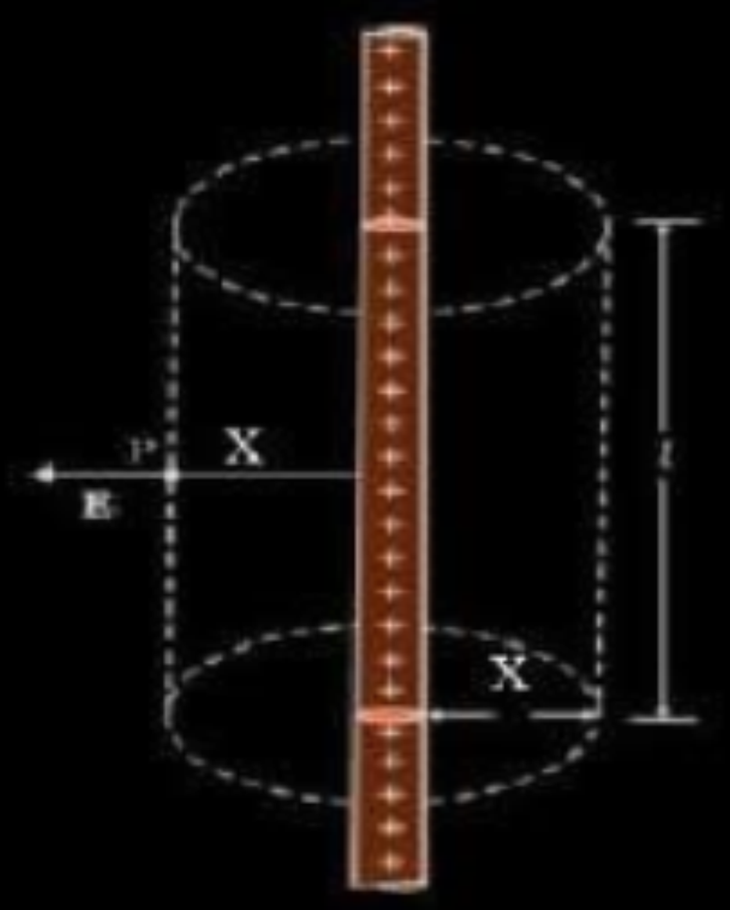
$$E = \frac{1}{4\pi\epsilon_0}\frac{Q-q}{x^2}$$ ½

(a) An infinitely long thin straight wire has a uniform linear charge density λ. Obtain the expression for the electric field (E) at a point lying at a distance x from the wire, using Gauss' law.

(b) Show graphically the variation of this electric field E as a function of distance x from the wire. **3**

(a)	Derivation for electric field due to a uniformly charged straight wire	2
(b)	Graph showing variation of electric field E vs distance x	1

(a)



½

$$\oint \vec{E}.\overrightarrow{dS} = \frac{q}{\epsilon_0}$$

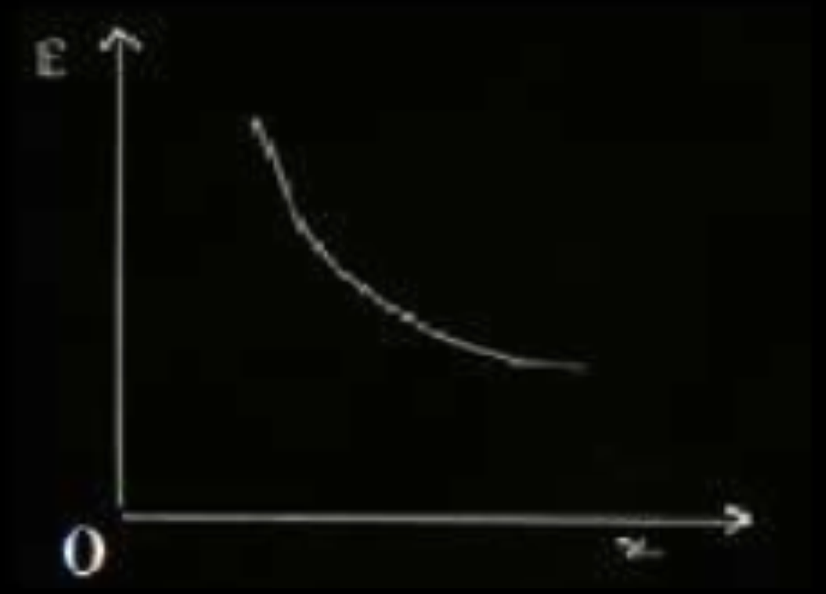
$$\int \vec{E}\,.\overrightarrow{dS_1} + \int \vec{E}\,.\overrightarrow{dS_2} + \int \vec{E}\,.\overrightarrow{dS_3} = \frac{\lambda l}{\epsilon_0}$$

$$E\ dS_1 \cos 90^0 + EdS_2 \cos 90^0 + EdS_3 \cos 0^0 = \frac{\lambda l}{\epsilon_0}$$ ½

$$0 + 0 + E \times 2\pi x l = \frac{\lambda l}{\epsilon_0}$$ ½

$$E = \frac{\lambda}{2\pi\epsilon_0 x}$$ ½

(b)



1

2. The electric flux emerging out from 1 C charge is

(A) $\frac{1}{\varepsilon_0}$

(B) 4π

(C) $\frac{4\pi}{\varepsilon_0}$

(D) ε_0

2. The electric flux emerging out from 1 C charge is

(A) $\frac{1}{\varepsilon_0}$

(B) 4π

(C) $\frac{4\pi}{\varepsilon_0}$

(D) ε_0

(A)

$\frac{1}{\epsilon_o}$

11. If the electric flux entering and leaving a closed surface in air are ϕ_1 and ϕ_2 respectively, the net electric charge enclosed within the surface is __________ . *1*

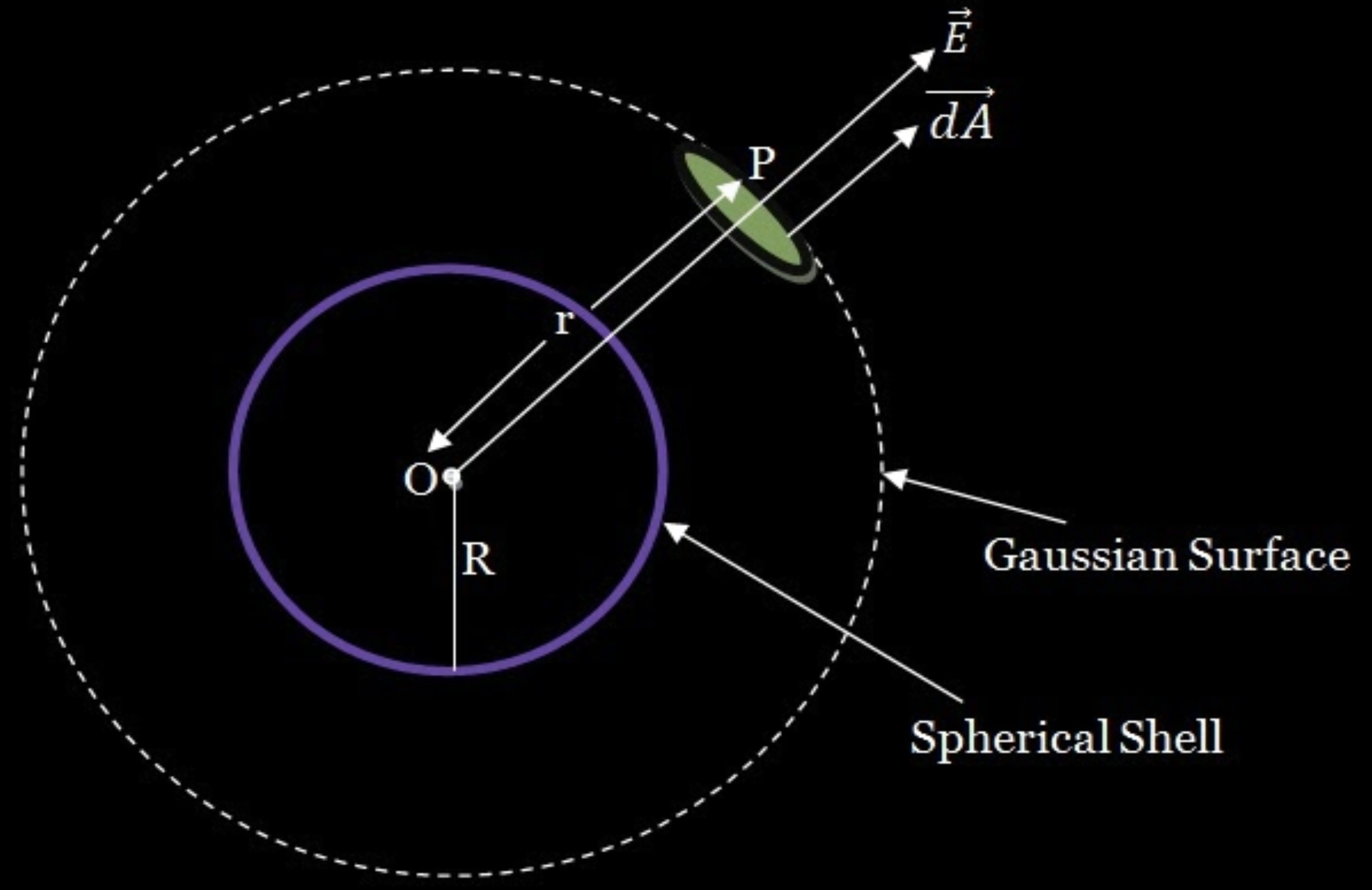
11. If the electric flux entering and leaving a closed surface in air are ϕ_1 and ϕ_2 respectively, the net electric charge enclosed within the surface is ___________ . *1*

$(\emptyset_2 - \emptyset_1)\varepsilon_0/\ (\emptyset_1 - \emptyset_2)\varepsilon_0$

35. (a) Use Gauss's law to show that due to a uniformly charged spherical shell of radius R, the electric field at any point situated outside the shell at a distance r from its centre is equal to the electric field at the same point, when the entire charge on the shell were concentrated at its centre. Also plot the graph showing the variation of electric field with r, for $r \leq R$ and $r \geq R$.

(b) Two point charges of $+1\ \mu C$ and $+4\ \mu C$ are kept 30 cm apart. How far from the $+1\ \mu C$ charge on the line joining the two charges, will the net electric field be zero ? *5*

(a) Expression for electric field outside a charged shell	2
Graph of E vs r	1
b) Location of point where field is zero	2



½

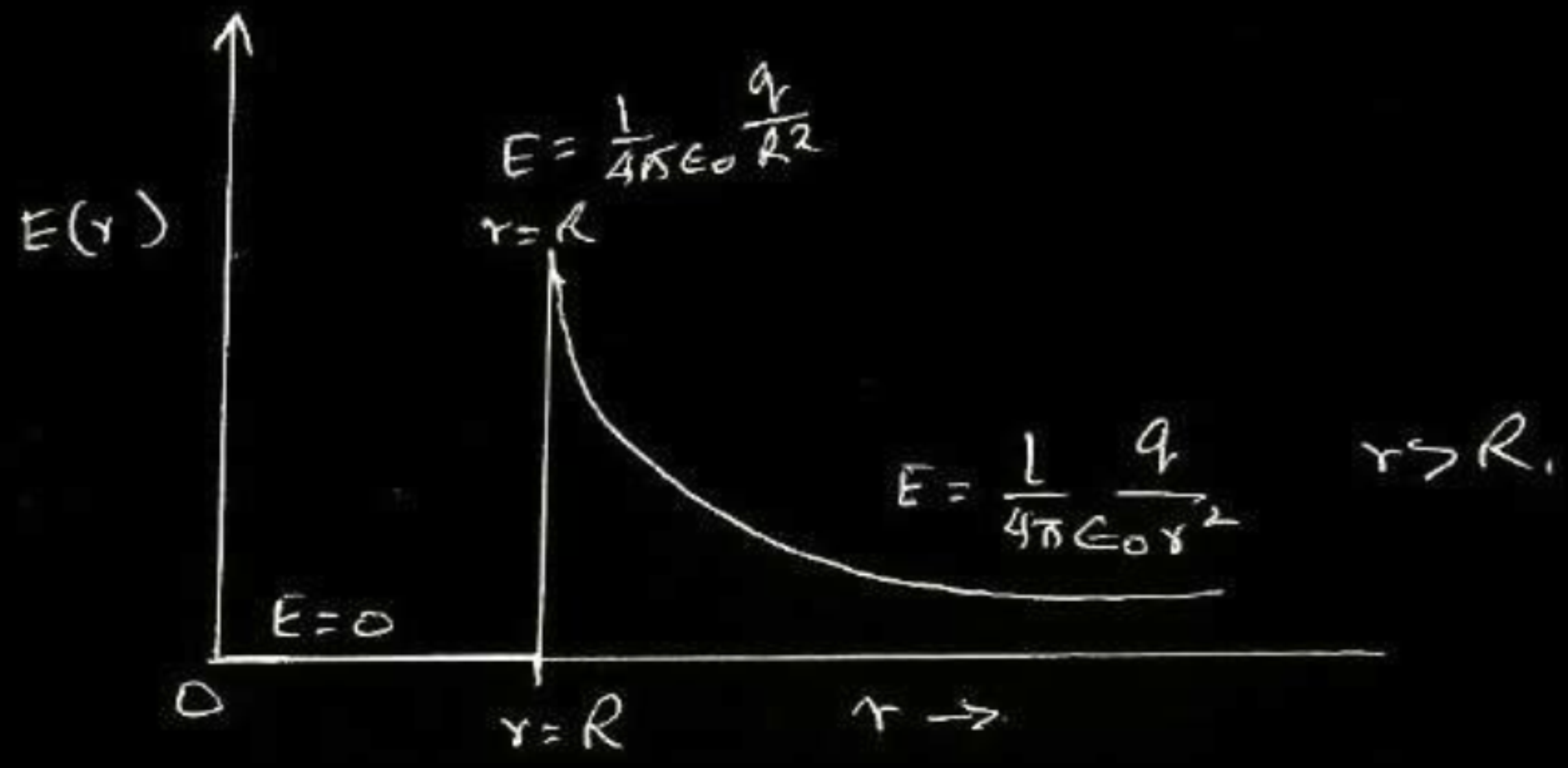
$$\phi = \frac{q}{\epsilon_0}$$ ½

$$E \times 4\pi r^2 = \frac{\sigma(4\pi R^2)}{\epsilon_0}$$ ½

$$E = \frac{1}{4\pi\epsilon_o}\frac{q}{r^2}$$ ½

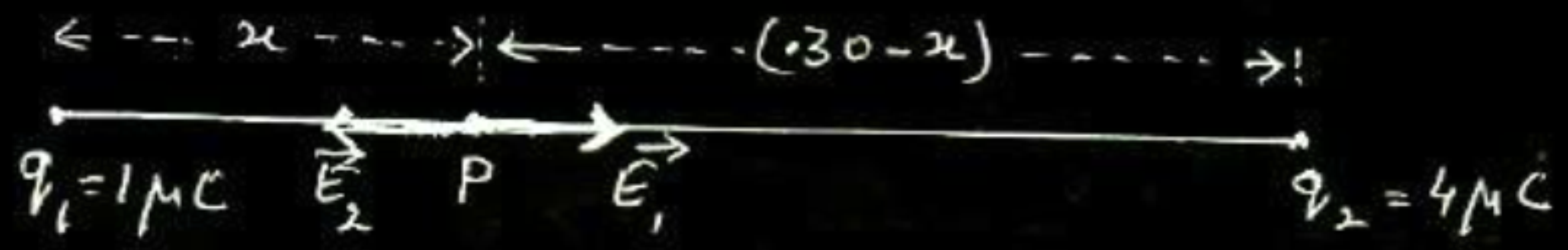
$$[\therefore q = \sigma(4\pi R^2)]$$

which is electric field due to a point charge q at a distance r from it



1

(b)



½

$E_1 = E_2$

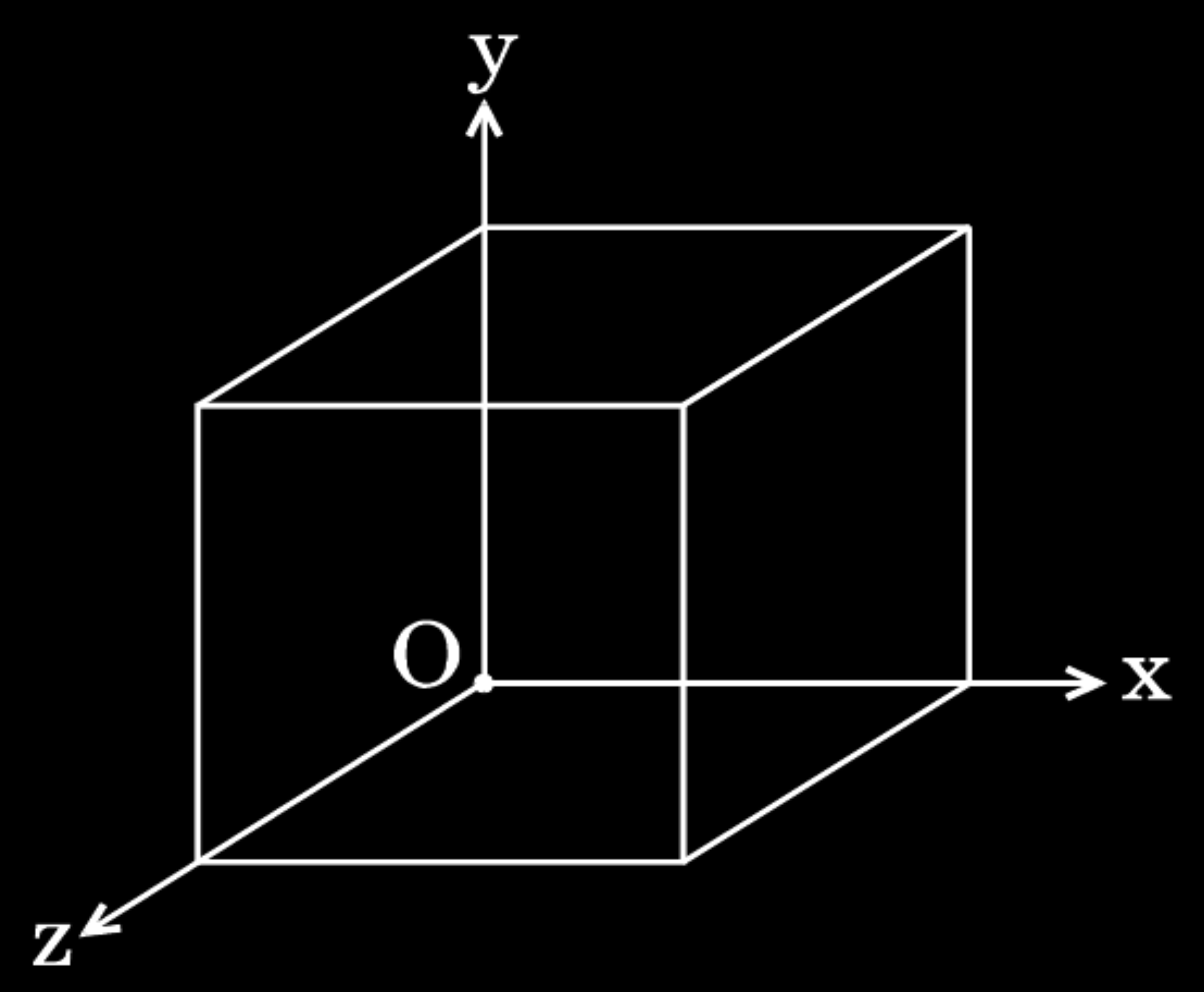
$$\frac{1}{4\pi\epsilon_o}\frac{1 \times 10^{-6}}{x^2} = \frac{1}{4\pi\epsilon_o}\frac{4 \times 10^{-6}}{(0.3 - x)^2}$$ ½

$$(0.3 - x)^2 = 4x^2$$ ½

$$0.3 - x = 2x$$

$$x = 0.1m = 10\ cm\ (to\ the\ right\ of\ q_1)$$ ½

(b) A cube of side 20 cm is kept in a region as shown in the figure. An electric field $\vec{E}$ exists in the region such that the potential at a point is given by V = 10x + 5, where V is in volt and x is in m.



Find the

(i) electric field $\vec{E}$, and

(ii) total electric flux through the cube.

(b)

(i)

$$E = \frac{-dV}{dx} = -\frac{d}{dx}(10x+5)$$ 1

$$\therefore \vec{E} = -10\hat{i}\ N/C$$ ½

(ii) Electric flux through the cube, ϕ =sum of electric flux through 6 faces

Electric flux through faces perpendicular Y and Z axis = 0 ½

$\because$ E is along x axis

Electric flux through faces perpendicular to x axis

$$= \phi_1 + \phi_2$$ ½

$$= 10 \times (0.2)^2 - 10 \times (0.2)^2$$

=0 ½

1. If the net electric flux through a closed surface is zero, then we can infer *1*

(A) no net charge is enclosed by the surface.

(B) uniform electric field exists within the surface.

(C) electric potential varies from point to point inside the surface.

(D) charge is present inside the surface.

1. If the net electric flux through a closed surface is zero, then we can infer *1*

(A) no net charge is enclosed by the surface.

(B) uniform electric field exists within the surface.

(C) electric potential varies from point to point inside the surface.

(D) charge is present inside the surface.

(A)

no net charge is enclosed by the surface

35. (a) Using Gauss law, derive expression for electric field due to a spherical shell of uniform charge distribution σ and radius R at a point lying at a distance x from the centre of shell, such that

(i) $0 < x < R$, and

(ii) $x > R$.

(b) An electric field is uniform and acts along + x direction in the region of positive x. It is also uniform with the same magnitude but acts in – x direction in the region of negative x. The value of the field is $E = 200$ N/C for $x > 0$ and $E = -200$ N/C for $x < 0$. A right circular cylinder of length 20 cm and radius 5 cm has its centre at the origin and its axis along the x-axis so that one flat face is at $x = +10$ cm and the other is at $x = -10$ cm.

Find :

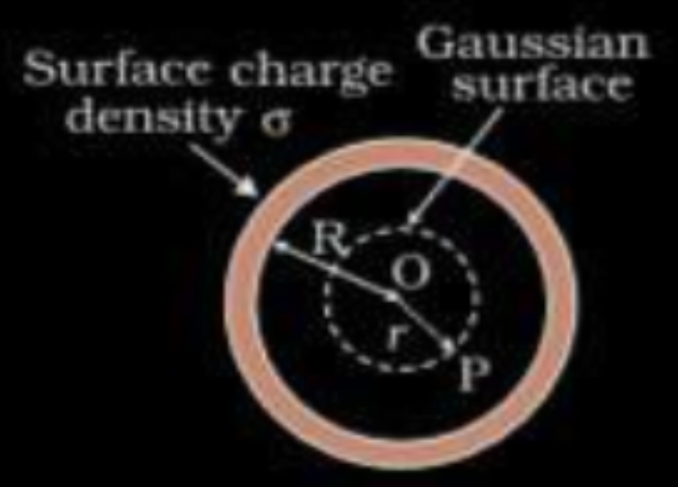
(i) The net outward flux through the cylinder.

(ii) The net charge present inside the cylinder. 5

(a) (i) Electric Field inside hollow sphere	1½ mark
(ii) Electric Field outside hollow sphere	1½ mark
(b) (i) The net outward flux through cylinder	1 mark
(ii) The net charge present inside the cylinder	1 mark

(a)

(i)



½

According to Gauss's Law

$$\oint \vec{E}.\overrightarrow{dA} = \frac{q_{in}}{\epsilon_0}$$ ½

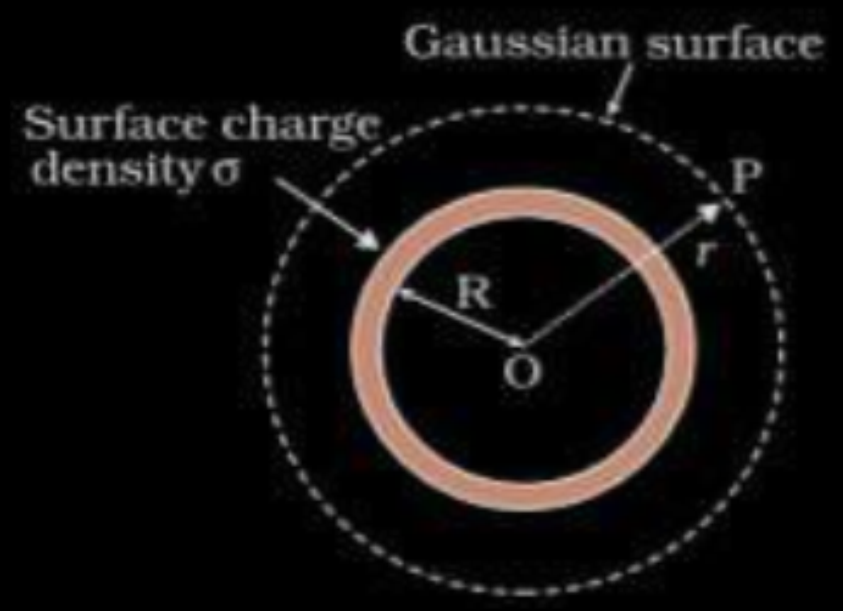
$\because$ *inside hollow sphere*

$q_{in} = 0$

$$\therefore \oint \vec{E}\,.\overrightarrow{dA} = 0$$

$E = 0$ ½

(ii)



½

$$q = \sigma 4\pi R^2$$

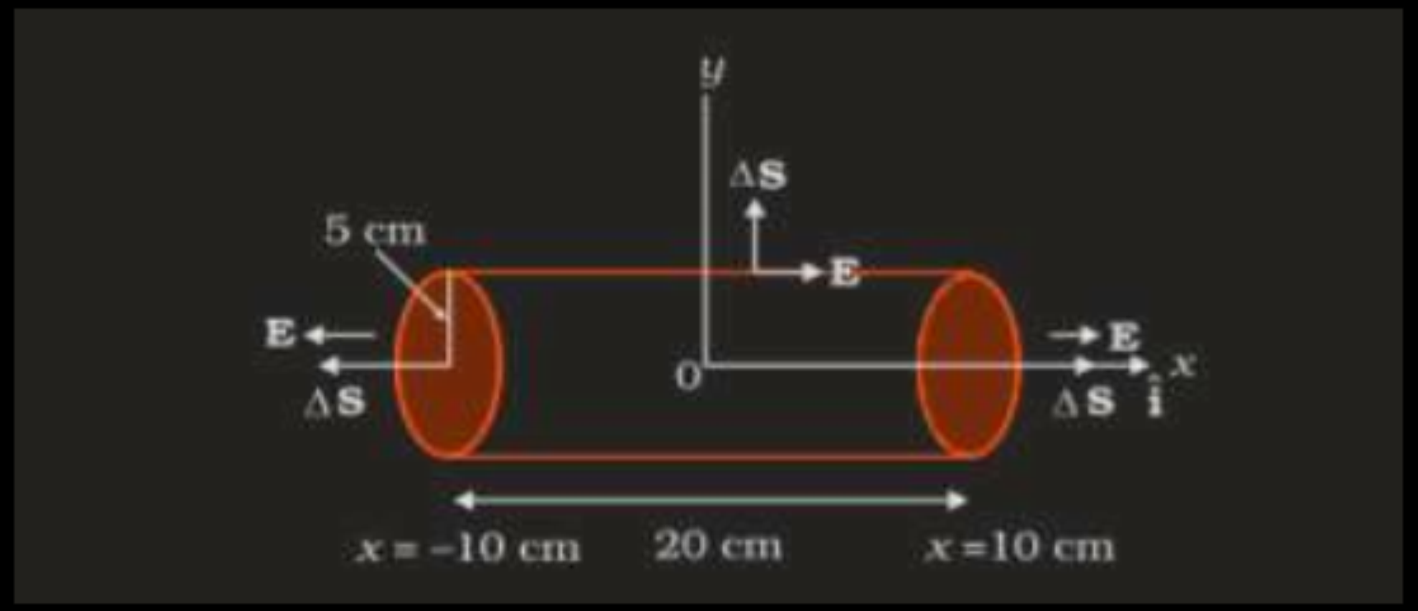
$$\oint \vec{E}.\overrightarrow{dA} = \frac{q}{\epsilon_0}$$ ½

$$E \oint dA = \frac{q}{\epsilon_0}$$

$$E.4\pi x^2 = \frac{\sigma 4\pi R^2}{\epsilon_0}$$

$$E = \frac{\sigma R^2}{\epsilon_0 x}$$ ½

b)



(i) The net outward flux through cylinder

$$\phi = EA + EA = 2EA \qquad A = \pi r^2$$

$= 2 \times 200 \times 3.14 \times 0.05 \times 0.05$ ½

$$= 3.14 \frac{N}{C} m^2$$ ½

(ii) The net charge present inside the cylinder

$$q = \epsilon_0 \phi$$ ½

$q = 8.854 \times 3.14 \times 10^{-12}$ ½

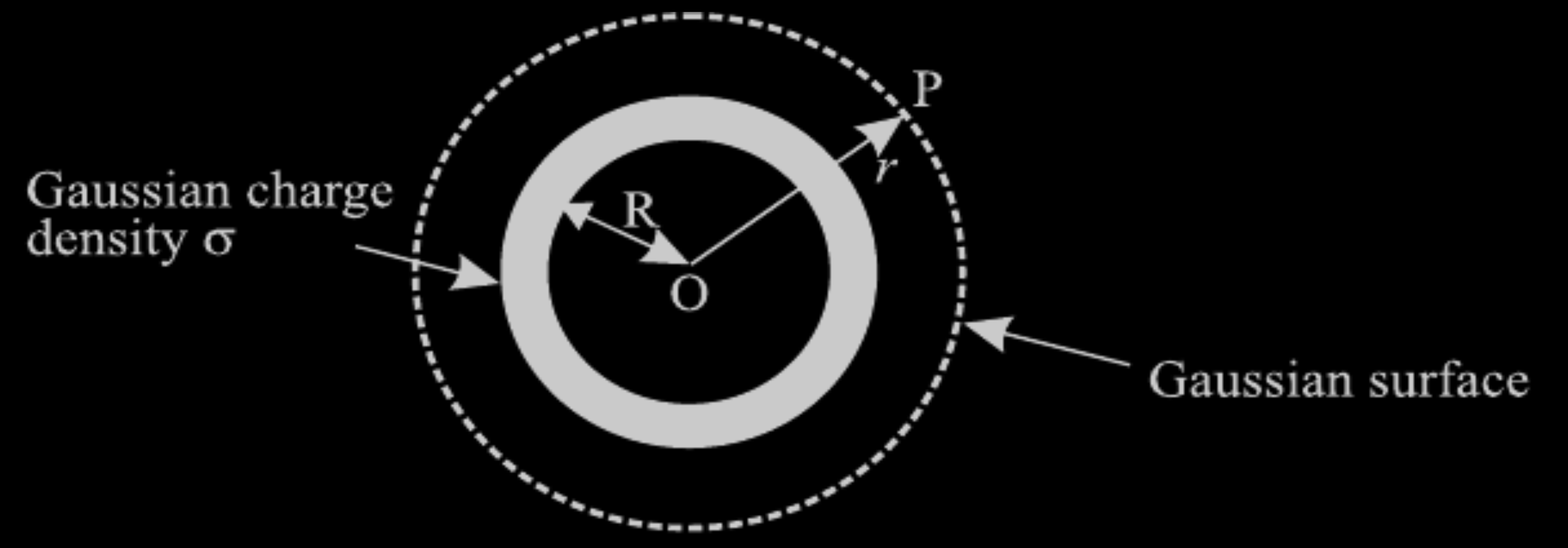
$= 2.78 \times 10^{-11}$ C

Apply Gauss's law to show that for a charged spherical shell, the electric field outside the shell is, as if the entire charge were concentrated at the centre.

OR

Two large parallel plane sheets have uniform charge densities $+\sigma$ and $-\sigma$. Determine the electric field (i) between the sheets, and (ii) outside the sheets.

i) Labelled diagram	½
ii) Flux through Gaussian surface	½
iii) Calculation & Result	½+½



½

Flux through the small section of Gaussian surface

$$\phi = \oint \vec{E}.\vec{ds}$$

$$\therefore \phi = \oint Eds\cos\theta$$

$$\because E \parallel \vec{ds},\ \theta = 0$$

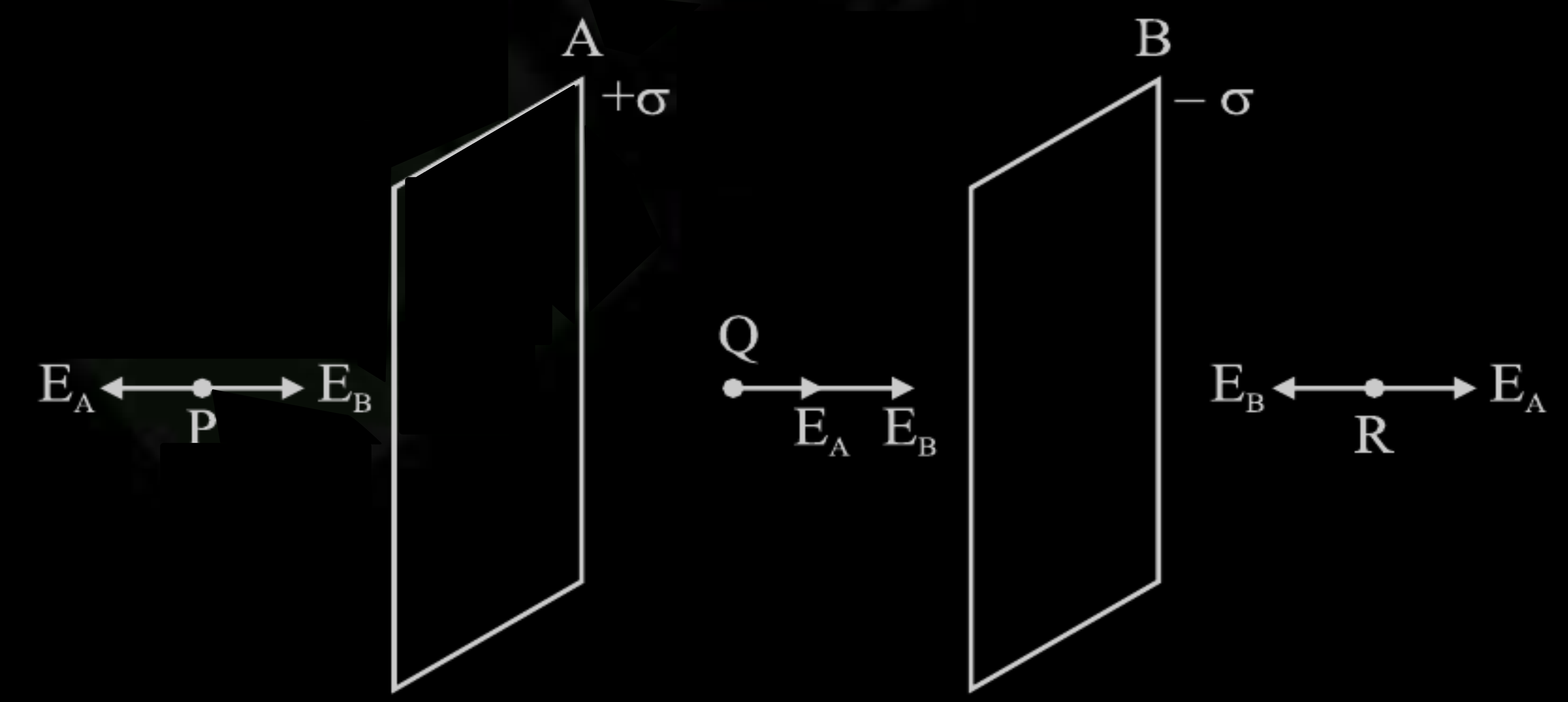
$$\therefore \phi = E.4\pi R^2 \quad \text{.......................} (1)$$ ½

Applying Gauss's theorem

$$\phi = \frac{q}{\varepsilon_0} \quad \text{.......................} (2)$$

from equations 1 and 2

$$E = \frac{1}{4\pi\varepsilon_0}.\frac{q}{R^2}$$ ½



½

Now Electric field Intensity due to a plane sheet of charge

$$E = \frac{\sigma}{2\varepsilon_0}$$ ½

Here

$$E_A = \frac{+\sigma}{2\varepsilon_0} \text{ and } E_B = \frac{-\sigma}{2\varepsilon_0}$$

(i) Electric field at Point Q (In between the sheets)

$$\vec{E} = \vec{E}_A + \vec{E}_B = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$$ ½

(ii) Field at the point P or R

$$\vec{E} = \vec{E}_A + \vec{E}_B = \frac{\sigma}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0} = 0$$ ½

Two identical conducting balls A and B have charges –Q and +3Q respectively. They are brought in contact with each other and then separated by a distance d apart. Find the nature of the Coulomb force between them.

OR

A metallic spherical shell has an inner radius R_1 and outer radius R_2. A charge Q is placed at the centre of the shell. What will be the surface charge density on the (i) inner surface, and (ii) outer surface of the shell ?

Two identical conducting balls A and B have charges –Q and +3Q respectively. They are brought in contact with each other and then separated by a distance d apart. Find the nature of the Coulomb force between them.

OR

A metallic spherical shell has an inner radius R_1 and outer radius R_2. A charge Q is placed at the centre of the shell. What will be the surface charge density on the (i) inner surface, and (ii) outer surface of the shell ?

Repulsive	1
OR	
Surface charge density on inner surface $= -\dfrac{Q}{4\pi R_1^2}$	½
Surface charge density on Outer surface $= +\dfrac{Q}{4\pi R_2^2}$	½

Two large charged plane sheets of charge densities σ and -2σ C/m^2 are arranged vertically with a separation of d between them. Deduce expressions for the electric field at points (i) to the left of the first sheet, (ii) to the right of the second sheet, and (iii) between the two sheets.

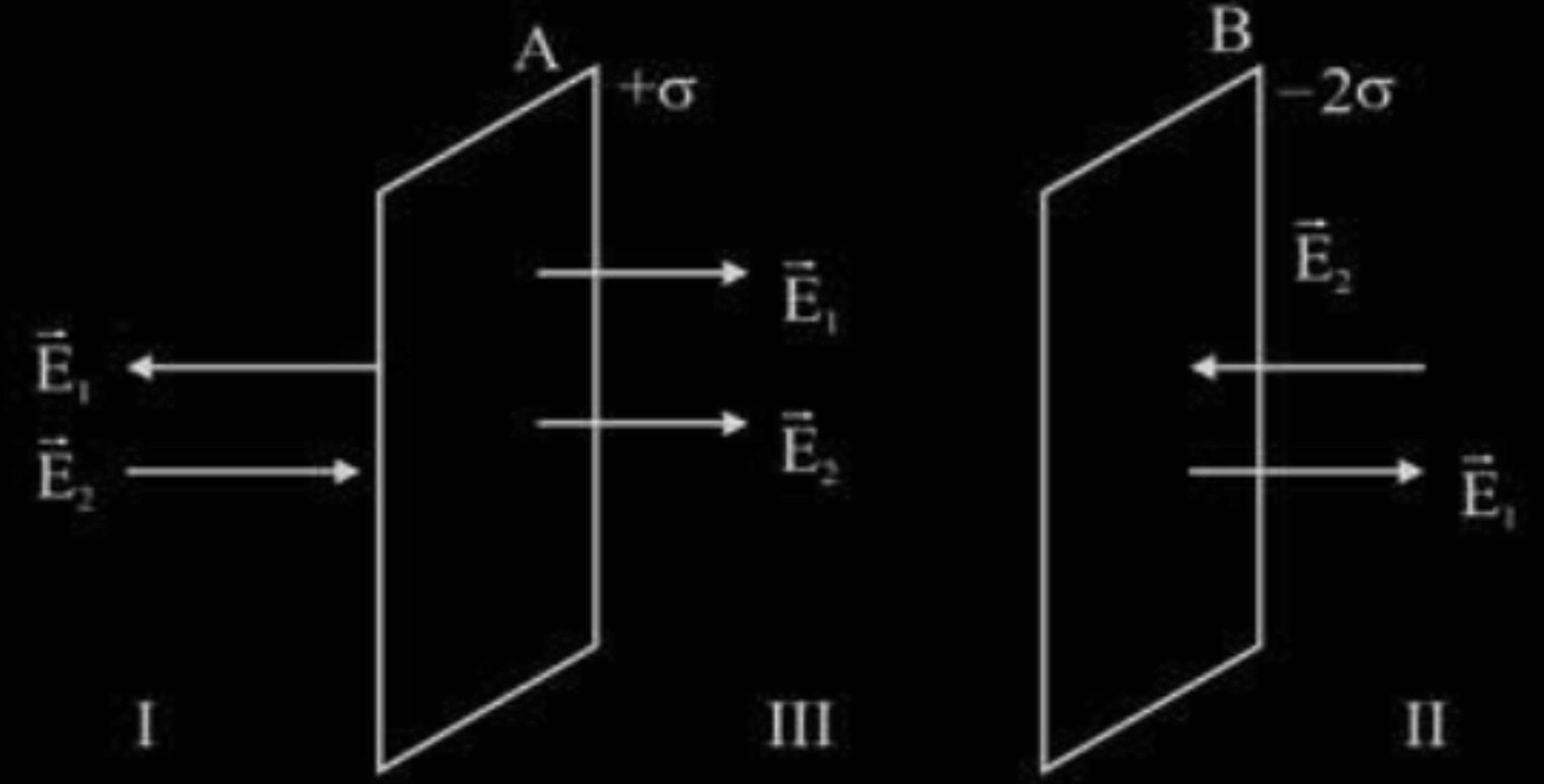
OR

A spherical conducting shell of inner radius r_1 and outer radius r_2 has a charge Q.

(a) A charge q is placed at the centre of the shell. Find out the surface charge density on the inner and outer surfaces of the shell.

(b) Is the electric field inside a cavity (with no charge) zero; independent of the fact whether the shell is spherical or not ? Explain.

Diagram



1½

Electric field in the region left of first sheet

$E_I = E_1 + E_2$

$E_I = \frac{\sigma}{\varepsilon_0} - \frac{\sigma}{2\varepsilon_0}$

$E_I = +\frac{\sigma}{2\varepsilon_0}$ ½

It is towards right

Electric field in the region to the right of second sheet

$E_{II} = \frac{\sigma}{2\varepsilon_0} - \frac{\sigma}{\varepsilon_0}$

$E_{II} = -\frac{\sigma}{2\varepsilon_0}$ ½

It is towards left

Electric field between the two sheets

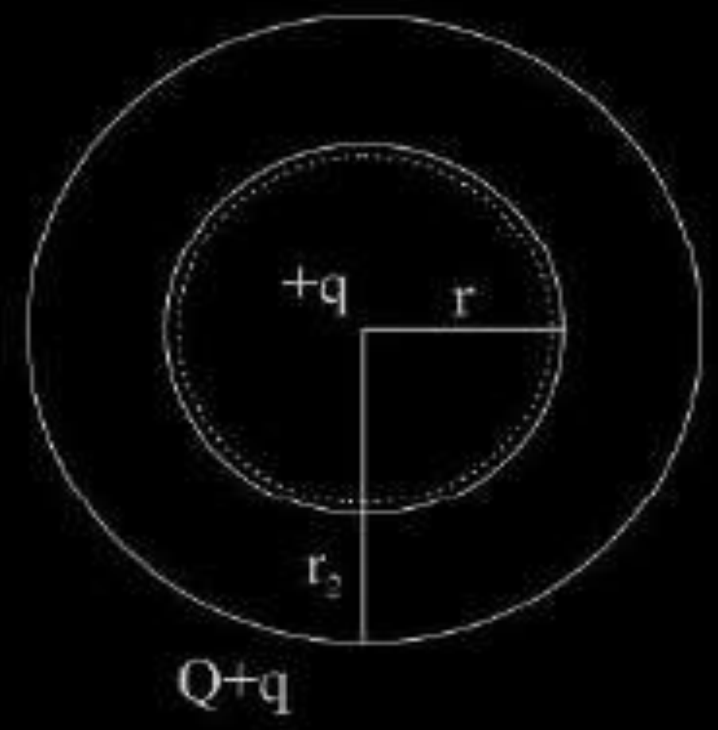
$E_{III} = E_1 + E_2$

$E_{III} = \frac{\sigma}{\varepsilon_0} + \frac{\sigma}{2\varepsilon_0}$

$E_{III} = \frac{3\sigma}{2\varepsilon_0}$ ½

Electric field is towards the right

(a) Diagram



½

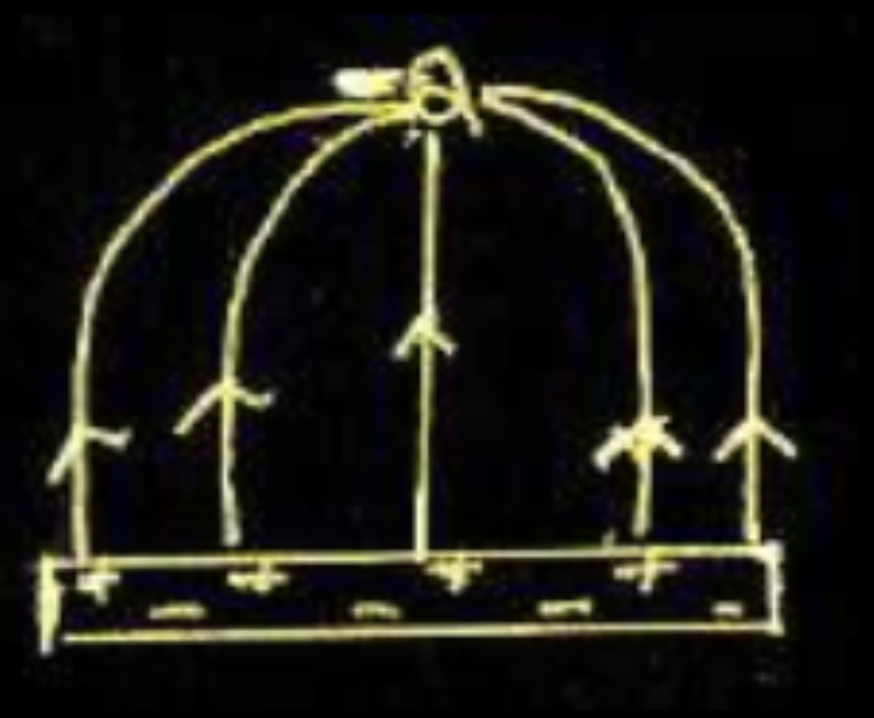
The surface charge density on inner surface of the shell is $\sigma_1 = -\frac{q}{4\pi r_1^2}$ 1

The surface charge density on outer shell is $\sigma_2 = \frac{Q+q}{4\pi r_2^2}$ ½

(b) Consider a Gaussian surface inside the shell, net flux is zero since $q_{net} = 0$. According to Gauss's law it is independent of shape and size of shell. 1

Draw the pattern of electric field lines, when a point charge – Q is kept near an uncharged conducting plate.

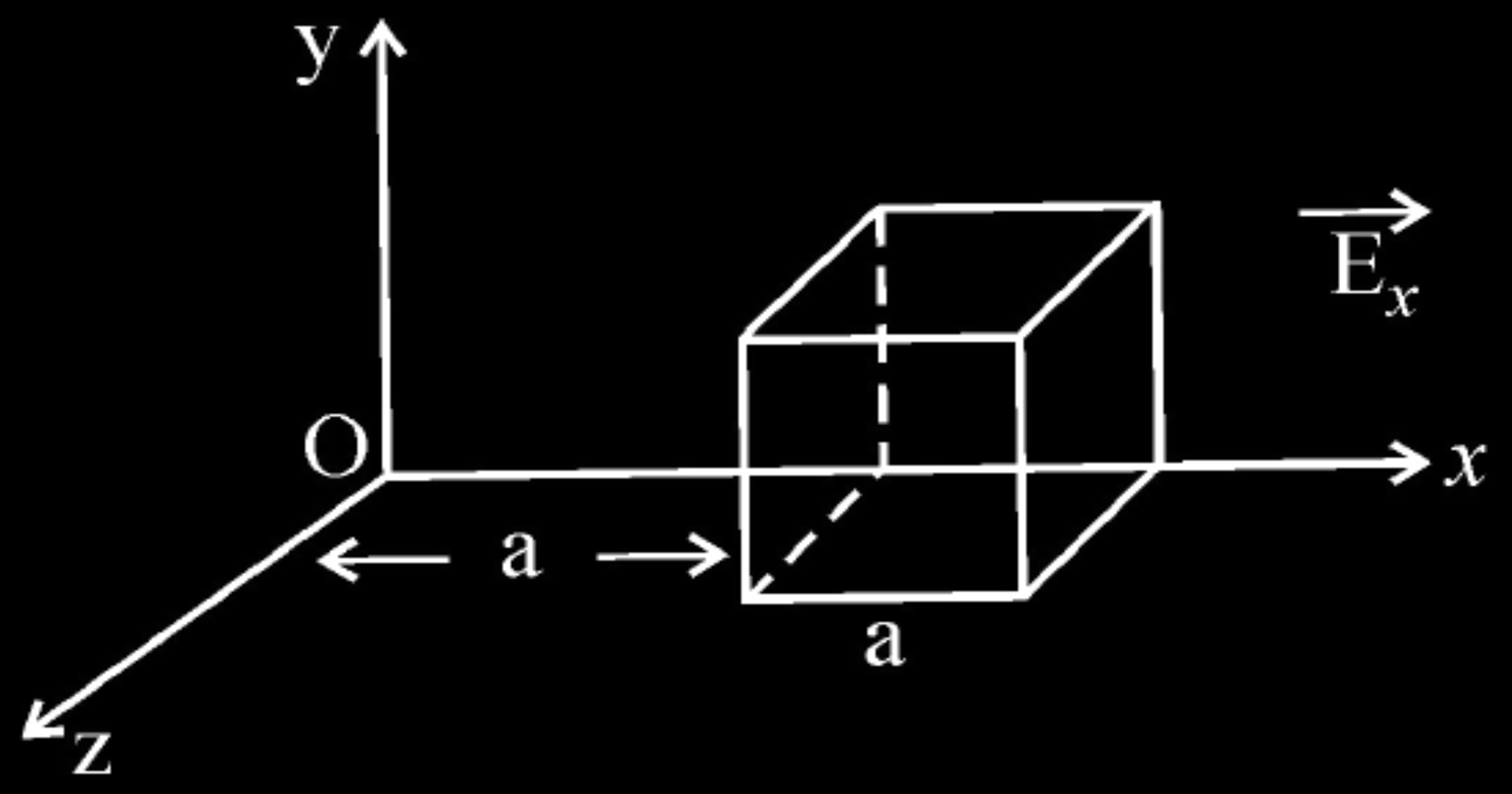
Draw the pattern of electric field lines, when a point charge – Q is kept near an uncharged conducting plate.



1

[Note: i) Deduct ½ mark, if arrows are not shown.
ii) do not deduct any mark, if charges on the plates are not shown]

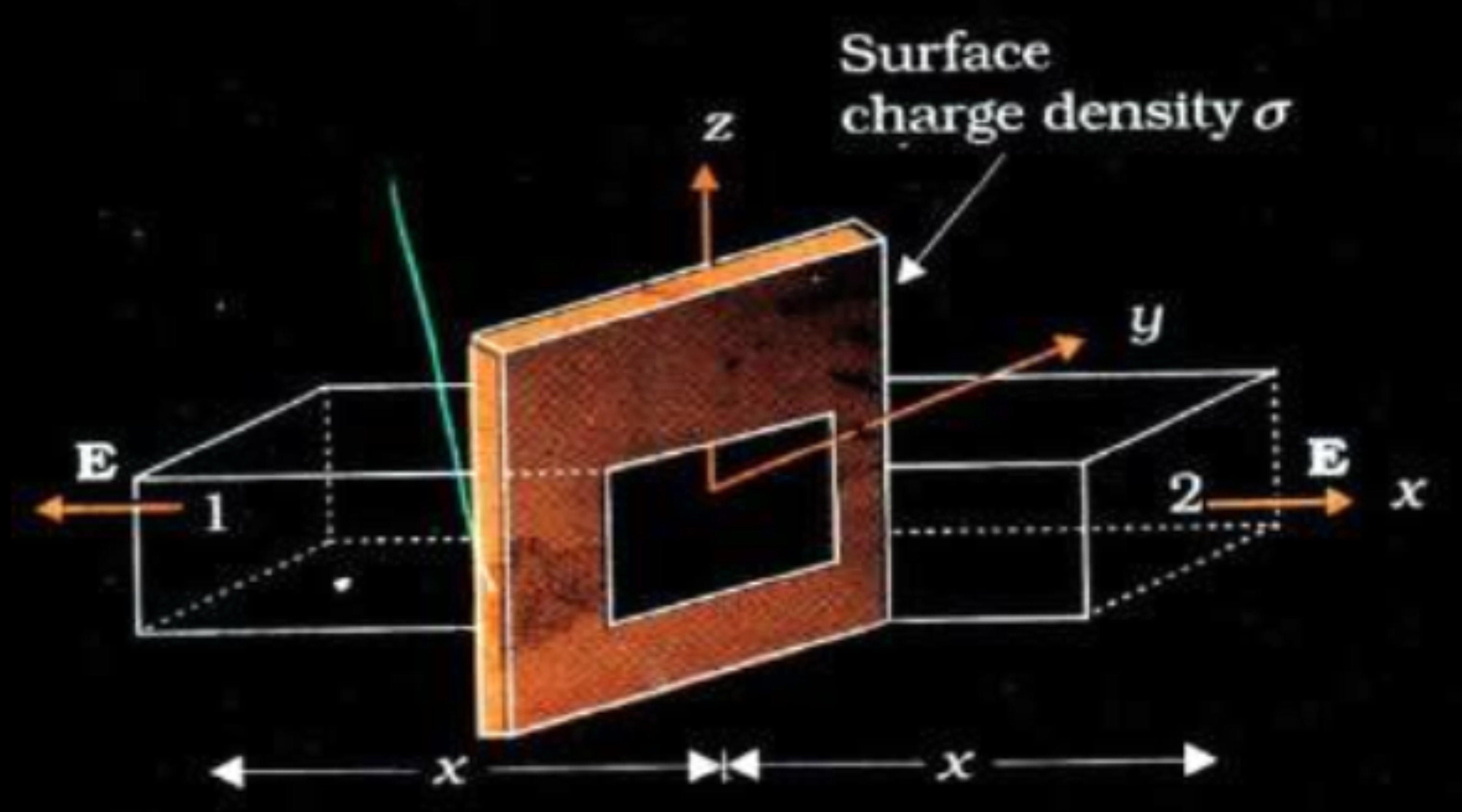
Define electric flux and write its SI unit. The electric field components in the figure shown are : $E_x = \alpha x$, $E_y = 0$, $E_z = 0$ where $\alpha = \frac{100\ \text{N}}{\text{Cm}}$. Calculate the charge within the cube, assuming a = 0.1m.



(a) Use Gauss's theorem to find the electric field due to a uniformly charged infinitely large plane thin sheet with surface charge density σ.

(b) An infinitely large thin plane sheet has a uniform surface charge density $+\sigma$. Obtain the expression for the amount of work done in bringing a point charge q from infinity to a point, distant r, in front of the charged plane sheet.

a)



$$\oint E.ds = \frac{q}{\varepsilon_0}$$ ½

The electric field E points outwards normal to the sheet. The field lines are parallel to the Gaussian surface except for surfaces 1 and 2. Hence the net flux $= \oint E.ds = EA + EA$ where A is the area of each of the surface 1 and 2. ½

1

$$\therefore \oint E.ds = \frac{q}{\varepsilon_0} = \frac{\sigma A}{\varepsilon_0} = 2EA;$$ **1**

$$E = \frac{\sigma}{2\varepsilon_0}$$

b)

$$W = q\int_{\infty}^{r} \vec{E}.d\vec{r}$$ ½

$$= q\int_{\infty}^{r} (-Edr)$$ ½

$$= -q\int_{\infty}^{r} \left(\frac{\sigma}{2\epsilon_0}\right) dr$$ ½

$$= \frac{q\sigma}{2\in}|\infty - r|$$

$$\Longrightarrow (\infty)$$ ½

How does the electric flux due to a point charge enclosed by a spherical Gaussian surface get affected when its radius is increased ?

How does the electric flux due to a point charge enclosed by a spherical Gaussian surface get affected when its radius is increased ?

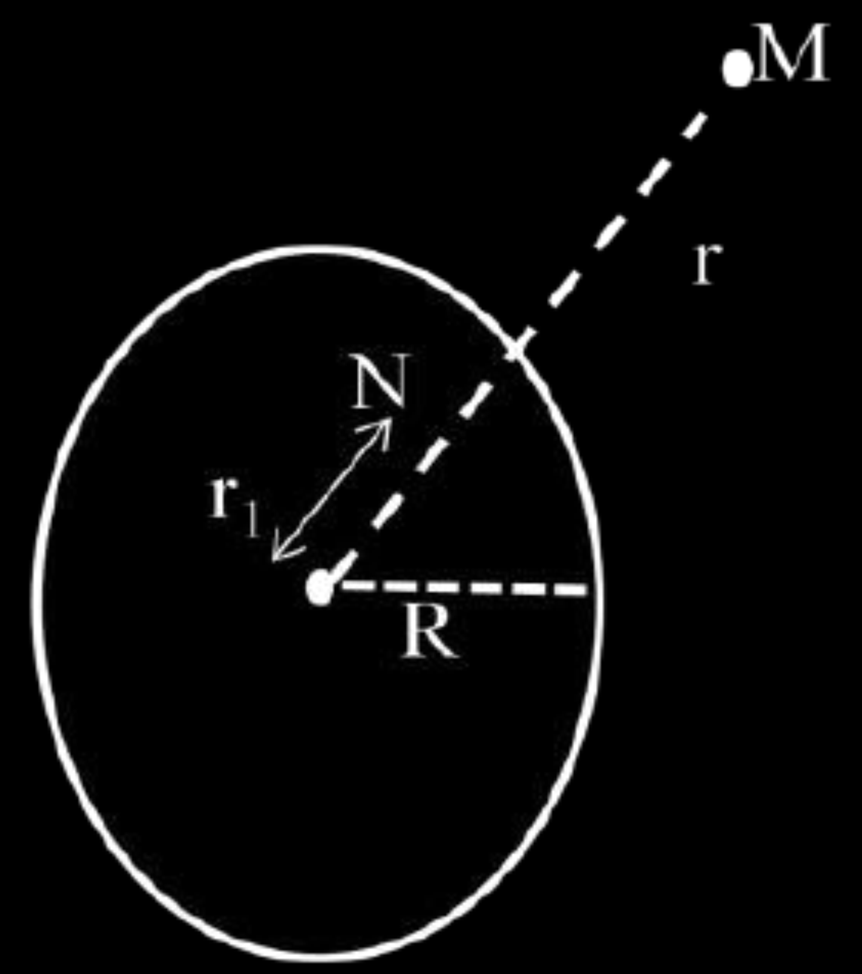
Electric flux remains unaffected.

Find the electric field intensity due to a uniformly charged spherical shell at a point (i) outside the shell and (ii) inside the shell. Plot the graph of electric field with distance from the centre of the shell.

We have by Gauss's law $\oint \vec{E}.\overrightarrow{dS} = \frac{Q_{enclosed}}{\epsilon_0}$ ½

Let Q be the total charge on the shell

(i) For the point M outside the shell, we have



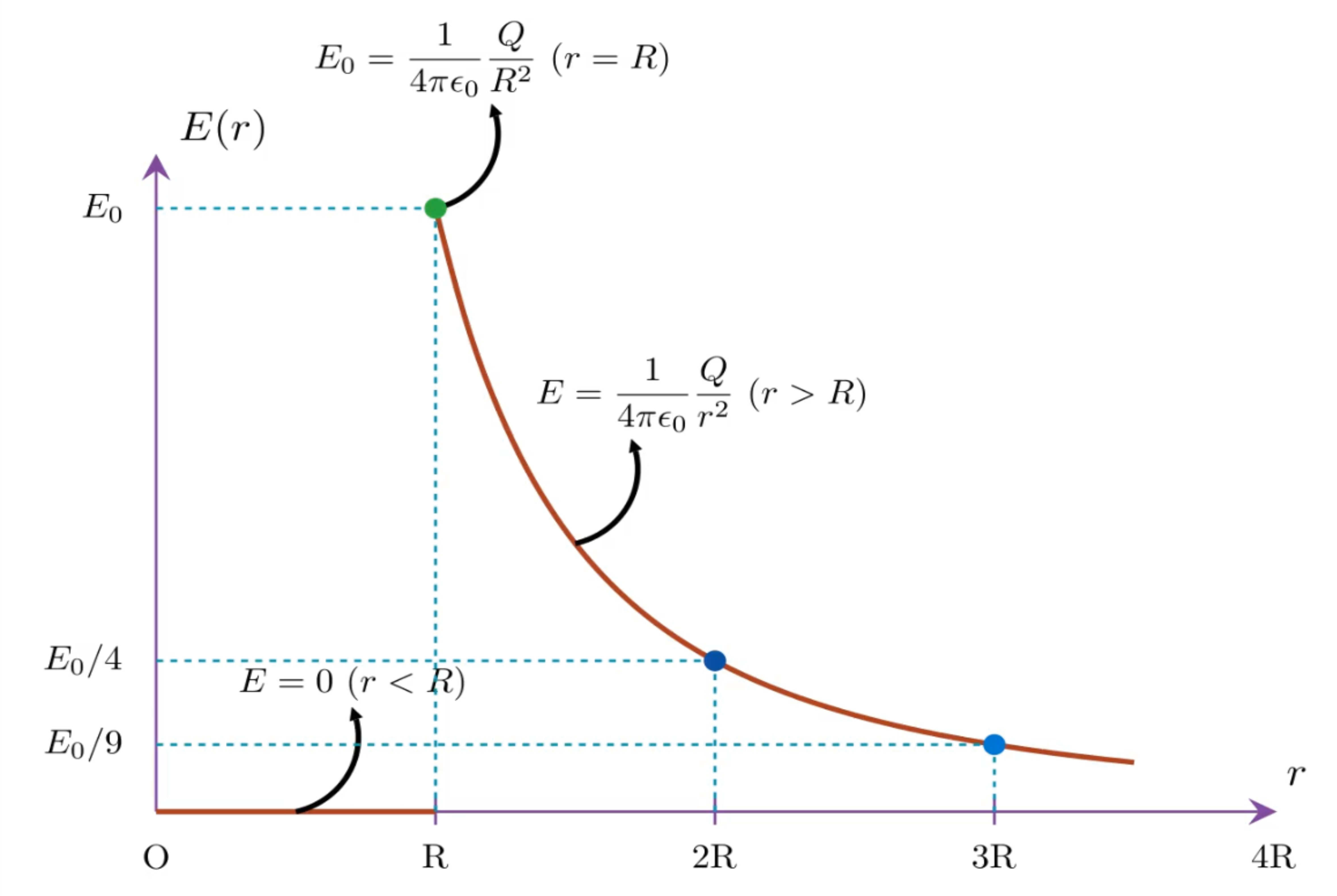
$$E.4\pi r^2 = \frac{Q}{\epsilon_0}$$ ½

$$\therefore E = \frac{1}{4\pi\epsilon_o}\frac{Q}{r^2}$$ ½

(ii) For the point N inside the shell, as charge enclosed inside the shell is zero. ½

$$E.4\pi r_1^2 = 0$$ ½

$$\therefore E = 0$$



Graph of Electric field (E) versus r for spherical shell.